

SINOPSIS KURSUS/ COURSE SYNOPSIS

Jabatan Dietetik/ Department of Dietetics

DTK3105 Prinsip Penyediaan dan Keselamatan Makanan / *Principles of Food Preparation and Safety* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi pengenalan kepada prinsip penyediaan makanan dengan mengambil kira aspek keselamatan makanan. Kesan perubahan fizikal, kimia dan nilai pemakanan pada makanan yang dimasak turut diaplikasikan. Aspek kelestarian makanan turut dibincangkan.

This course includes an introduction to food preparation principles, taking into account food safety aspects. The effects of physical, chemical, and nutritional changes in cooked food are also applied. Aspects of food sustainability are also discussed.

DTK3106 Pengurusan Perkhidmatan Makanan Institusi I / *Institutional Foodservice Management I* 3(3+0)

Prasyarat : DTK3105

Kursus ini meliputi sistem perkhidmatan makanan dan prinsip pengurusan perkhidmatan makanan institusi. Komponen penting dalam sistem perkhidmatan makanan bermula dari perancangan menu kepada perolehan barangan ditekankan. Aspek kelestarian dan peluang aktiviti keusahawanan dalam perkhidmatan makanan turut didedahkan.

This course is covering food service systems and principles of institutional food service management. Important components of the food service system start from menu planning to procurement of goods are emphasized. Sustainability aspects and entrepreneurial activity opportunities in food service are also highlighted.

DTK3221 Patofisiologi dan Pengurusan Klinikal Penyakit / *Pathophysiology and Clinical Management of Diseases* 4(4+0)

Prasyarat : Tiada

Kursus ini meliputi aspek patofisiologi penyakit yang berkaitan pemakanan. Ia juga merangkumi interpretasi aspek biokimia dan klinikal penyakit, disusuli dengan diagnosis, pengurusan perubatan dan farmakologi penyakit berkaitan pemakanan. Aplikasi dietetik dalam imunologi dan interaksi ubat-nutrien turut dibincangkan.

This course covers aspects of the pathophysiology of nutrition-related diseases. It also includes the interpretation of the biochemical and clinical aspects of these diseases, followed by diagnosis, medical management, and pharmacology of nutrition-related diseases. The application of dietetics in immunology and drug-nutrient interactions is also discussed.

DTK3222 Penilaian Taraf Pemakanan dan Kesihatan Pesakit / *Assessment of Nutrition and Health Status of Patients* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi penilaian pemakanan yang merupakan sebahagian daripada proses penjagaan pemakanan bagi menentukan taraf kesihatan pesakit. Pelbagai kaedah penilaian taraf pemakanan dan kesihatan pesakit turut ditekankan. Data penilaian pemakanan pesakit dibandingkan dengan piawai.

This course encompasses nutritional assessment, which is part of the nutritional care process to determine the health status of patients. Various methods of assessing nutritional status and patient health are also emphasized. Patient nutritional assessment data is compared with standards.

DTK3231 Pengalaman Awal Klinik Dietetik I / *Early Dietetics Clinical Experience I* 1(0+1)

Prasyarat : Tiada

Kursus ini merangkumi pendedahan awal tentang sistem penjagaan kesihatan dan perkhidmatan dietetik. Pelajar didedahkan dengan perkhidmatan dietetik klinikal, perkhidmatan dietetik komuniti, pengurusan perkhidmatan makanan di institusi dan peranan pentadbiran di peringkat kementerian. Peranan ahli dietetik sebagai ahli profesional kesihatan turut dibincangkan.

This course covers an early exposure to healthcare systems and dietetic services. Students are introduced to clinical dietetic services, community dietetic services, food service management in institutions and administrative role in the ministry. The role of dietitians as healthcare professionals is also discussed.

DTK3232 Pengalaman Awal Klinik Dietetik II / *Early Dietetics Clinical Experience II* 1(0+1)

Prasyarat : DTK3231

Kursus ini adalah kursus berasaskan pengalaman yang memberi pendedahan kepada perkhidmatan dietetik klinikal melalui aplikasi Proses Penjagaan Pemakanan. Penekanan diberikan kepada amalan tingkah laku profesional, etika dan adab dalam mendekati pesakit. Teknik kaunseling individu dan kelompok dibincangkan dalam perbincangan kumpulan

This course is an experience-based course that provides exposure to clinical dietetic services through the application of Nutrition Care Process. Emphasis is given to practice of professional behavior, ethics and manners in approaching patients. Individual and group counseling techniques are discussed in group discussion.

DTK3003 Proses Penjagaan Pemakanan / *Nutrition Care Process* 2(2+0)

Prasyarat : Tiada

Kursus ini memfokuskan kepada konsep dan prinsip NCP serta model, dalam penjagaan pemakanan pesakit. Ia secara khusus memberi tumpuan kepada terminologi piawai yang digunakan untuk mengenal pasti, merancang, dan memenuhi keperluan pemakanan pesakit. Penekanan juga diberikan kepada dokumentasi Proses Penjagaan Pemakanan melalui analisis kajian kes.

This course focuses on the concepts and principles of the Nutrition Care Process and model, in providing nutrition care to patients. It specifically emphasises standardised terminology to identify, plan for, and meet the nutritional needs of patients. The documentation of the Nutrition Care Process through case study analysis is also emphasised.

DTK3004 Seminar Dietetik / *Dietetics Seminar* 1(0+1)

Prasyarat : Tiada

Kursus ini meliputi penyediaan dan pengorganisasian maklumat saintifik berasaskan penyelidikan dan perkembangan dalam bidang dietetik. Pengulasan kritis dan perbincangan ke atas penulisan terkini yang diterbitkan dalam jurnal mengenai bidang yang spesifik dalam dietetik ditekankan.

This course covers the preparation and organisation of scientific information based on research and development in the field of dietetics. Critical review and discussion of recent publications in journals specific to the field of dietetics are emphasised.

DTK3107 Pengurusan Perkhidmatan Makanan Institusi II / *Institutional Foodservice Management II* 3(3+0)

Prasyarat : DTK3106

Kursus ini menekankan latihan amali pengeluaran makanan berkuantiti yang berkualiti. Pengawasan kos dan bahan dalam menjana pendapatan turut ditekankan. Sistem perkhidmatan makanan dan amalan lestari di pelbagai institusi turut didedahkan.

This course emphasizes practical training in producing quantity and quality food. Cost and ingredient control in generating income are also emphasized. Food service systems and sustainable practices in various institutions are also exposed.

DTK3224 Terapi Pemakanan Perubatan I / *Medical Nutrition Therapy I* 3(2+1)

Prasyarat : DTK3221

Kursus ini merupakan pengenalan kepada terapi pemakanan perubatan. Penekanan diberi kepada hubungkait antara penyakit dengan prinsip modifikasi diet terapeutik sebagai komponen penting dalam Proses Penjagaan Pemakanan bagi jagaan kesihatan primer. Kajian kes yang merangkumi perancangan, penyediaan dan penilaian diet terapeutik turut dijalankan.

This course is an introduction to medical nutrition therapy. It focuses on the relationship between diseases and the principles of therapeutic dietary modification as an essential component of Nutrition Care Process of primary health care. Case study that includes planning, preparation and evaluation of therapeutic diets is conducted.

DTK4959 Disertasi Bachelo / *Bachelor Dissertation* 6 (0+6)

Prasyarat : Tiada

Kursus ini merangkumi perancangan penyelidikan, pengumpulan, analisis dan tafsiran data, penyediaan dan pembentangan tesis. Penekanan diberi kepada kaedah saintifik dan etika profesional dalam penyelidikan.

This course includes research planning, data collection, analysis and interpretation, preparation and presentation of a thesis. Emphasis is given to scientific method and professional ethics in research

DTK3223 Kaunseling Pemakanan dan Pendidikan Pesakit / *Nutrition Counseling and Patient Education* 3(2+1)

Prasyarat : Tiada

Kursus ini merangkumi prinsip, teori kaunseling, komunikasi berkesan, dan kemahiran kaunseling dalam konteks amalan dietetik. Kaunseling individu dan kelompok dijalankan berdasarkan model proses penjagaan pemakanan. Teknik dalam pendidikan pesakit dititikberatkan. Kursus ini juga membincangkan etika dan tingkah laku profesional dalam kaunseling pemakanan dan pendidikan pesakit.

This course encompasses principles, counseling theory, effective communication, and counseling skills in the context of dietetic practice. Individual and group counseling are conducted based on the nutrition care process model. Techniques in patient education are emphasized. The course also discusses ethics and professional behavior in nutritionncounseling and patient education.

DTK3225 Terapi Pemakanan Perubatan II / *Medical Nutrition Therapy II* 3(2+1)

Prasyarat : DTK3224

Kursus ini memfokuskan penyakit yang lebih kompleks dengan menerapkan prinsip modifikasi diet terapeutik, mengambil kira status keterjaminan makanan pesakit dalam komponen pengawasan dan pengurusan penyakit. Kajian kes berdasarkan Proses Penjagaan Pemakanan turut ditekankan.

This course focuses on complex diseases that incorporate the principles of therapeutic diet modifications, considering patient's food security status as part of the control and management of the diseases. Case study based on the Nutrition Care Process is emphasized.

DTK3226 Terapi Pemakanan Perubatan III / *Medical Nutrition Therapy III* 3(2+1)

Prasyarat : DTK3225

Kursus ini memfokuskan kepada penjagaan pemakanan pesakit berkeperluan khusus dan kritikal dengan penerapan prinsip diet terapeutik termasuk pemberian sokongan pemakanan dalam komponen pengurusan penyakit. Kajian kes berdasarkan Proses Penjagaan Pemakanan turut ditekankan.

This course focuses on nutrition care of patients with special needs and critical incorporating the principles of therapeutic diets including nutrition support delivery. Case study based on the Nutrition Care Process is emphasized.

DTK3227 Pendekatan Kognitif dan Tingkah Laku dalam Amalan Dietetik / *Cognitive and Behavioural Approach in Dietetics Practice* 2(2+0)

Prasyarat : Tiada

Kursus ini merangkumi teori dan prinsip kognitif dan tingkahlaku dalam amalan dietetik. Teknik temu bual motivasi serta aplikasinya juga dibincangkan. Kajian kes yang menerapkan pendekatan kognitif dan tingkah laku yang berdasarkan bukti turut dijalankan.

This course covers the theories and principle of cognitive and behaviour in dietetic practice. Motivational interviewing techniques and its applications are also discussed. Case studies applying evidence-based approach of cognitive and behavioral are also conducted.

DTK3233 Pengalaman Awal Klinikal Dietetik III / *Early Dietetics Clinical Experience III* 1(0+1)

Prasyarat : DTK3232

Kursus ini adalah kursus berasaskan pengalaman awal klinikal yang menekankan kemahiran dalam Proses Penjagaan Pemakanan kepada pesakit. Faktor sosio-budaya juga diambil kira dalam strategi intervensi dan pengawasan tabiat pemakanan pesakit. Dokumentasi penjagaan pesakit turut dibincangkan

This course is an early clinical experience-based course that emphasizes skills in the Nutrition Care Process for patients. Socio-cultural factors are also taken into account in the intervention strategies and monitoring of patients' eating habits Patient care documentation is also discussed.

DTK3228 Pemakanan untuk Kecergasan, Sukan dan Rehabilitasi / *Nutrition For Fitness, Sport And Rehabilitation* 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi hubungan penting antara pemakanan dengan kecergasan, sukan serta proses rehabilitasi. Kesan langsung nutrien terhadap prestasi atletik dan pemulihan daripada kecederaan dalam aktiviti sukan akan dibincangkan. Penilaian kritikal dalam isu kontroversi berkaitan pemakanan dalam konteks kecergasan, sukan, rehabilitasi turut ditekankan.

This course covers the important relationship between nutrition with fitness, sports, and the rehabilitation process. The direct effects of nutrient manipulation on athletic performance and recovery from injuries in sports activities will be discussed. Critical assessment of controversial issues related to nutrition in the context of fitness, sports, and rehabilitation will also be emphasized.

DTK4911 Praktikum Profesional Dietetik Pengurusan Perkhidmatan 6(0+6)
Makanan Institusi / *Institutional Foodservice Management*
Professional Dietetics Practicum

Prasyarat : DTK3107

Kursus ini merangkumi latihan dan pengalaman bekerja di pelbagai operasi perkhidmatan institusi makanan. Kemahiran teknikal dan berfikir secara kritis terhadap operasi perkhidmatan makanan dan proses penjanaan pendapatan secara profesional dan beretika juga ditekankan.

This course encompasses training and work experience in various institutional food service operations. Technical skills and critical thinking on the operation of food services and income generation processes in a professional and ethical manner is also emphasized.

DTK4912 Praktikum Profesional Dietetik Klinikal / *Clinical Professional* 14(0+14)
Dietetics Practicum

Prasyarat : DTK3226

Kursus ini meliputi praktikum profesional dietetik klinikal pesakit luar dan pesakit dalam wad. Ia membentuk kemahiran terapi pemakanan perubatan termasuk saringan, penilaian status pemakanan, diagnosis, perancangan, intervensi, pemantauan dan dokumentasi Proses Penjagaan Pemakanan pesakit. Kemahiran dietetik dan pengembangan profesional turut ditekankan.

This course includes out-patient and in-patient clinical dietetic professional practicum. It develops the Medical Nutrition Therapy skills including screening, assessment, diagnosis, planning, intervention, monitoring and documentation of the patients' Nutrition Care Process. Dietetic skills and professional development are also emphasized.

DTK4913 Praktikum Profesional Dietetik Komuniti / *Community* 6(0+6)
Professional Dietetics Practicum

Prasyarat : Tiada

Kursus ini meliputi praktikum profesional dietetik komuniti yang menekankan kepada penilaian keperluan dan pengurusan program intervensi dietetik di peringkat komuniti. Kemahiran merancang, menjalankan dan menilai program intervensi dietetik komuniti turut ditekankan secara profesional.

This course covers the dietetic professional practicum which emphasizes the dietetic community needs assessment and program management of community dietetics interventions. Skills to design, conduct and evaluate community dietetic intervention program is also emphasized, professionally.

DTK4914 Praktikum Profesional Dietetik Elektif / *Elective Professional* 4(0+4)
Dietetics Practicum

Prasyarat : Tiada

Kursus ini merupakan praktikum profesional dietetik elektif di mana pelajar akan mendalami peranan ahli dietetik di pelbagai institusi pilihan di dalam atau di luar negara. Ia merangkumi sintesis pengalaman mengenai skop amalan dietetik di pelbagai lapangan termasuklah bidang dietetik memuncul. Aspek etika dan profesionalisme yang spesifik kepada skop amalan dietetik tersebut turut ditekankan.

This course is an elective professional dietetic practicum where students will delve into the role of dietitians in various institutions within or outside of the country. It encompasses the synthesis of experiences regarding the scope of dietetic practice in various fields, including emerging areas in dietetics. Specific aspects of ethics and professionalism related to the scope of dietetic practice are also emphasized.

DTK4001 Tajuk Khas Ddalam Dietetik / *Special Topics in Dietetics* 2(0+2)

Prasyarat : Tiada

Kursus ini mendedahkan pelajar terhadap teori atau amalan dalam bidang dietetik memuncul di bawah penyeliaan ahli akademik. Pelajar perlu mensintesis pengalaman melalui penglibatan dalam pengajian, projek, lawatan atau sangkutan. Aplikasi penggunaan teknologi digital dalam amalan dietetik memuncul turut ditekankan.

This course exposes students to theories and practices in the emerging field of dietetics under the supervision of academic advisors. Students are required to synthesize their experiences through engagement in studies, projects, field trips, and attachment. The application of digital technology in the emerging dietetic practices is also emphasized.

DTK3000 Kaunseling Pemakanan/ *Nutrition Counseling* 3 (2+1)

Prasyarat : PKK3208

Kursus ini merangkumi prinsip dan teori kaunseling, komunikasi berkesan, menemuduga dan kemahiran kaunseling. Penilaian sejarah diet dan status pemakanan dalam proses kaunseling turut ditekankan. Simulasi kaunseling individu dan kelompok bagi pencegahan penyakit dijalankan berdasarkan teori perubahan tingkahlaku. Kursus ini juga membincangkan etika dan tingkahlaku profesional dalam pemakanan

This course encompasses the principles and theories of counseling, effective communication, interviewing and counseling skills. Diet history and nutritional status assessment in counseling process are also emphasised. Simulations of individual and group counseling for the prevention of disease are conducted based on theory of behavioral change. This course also discusses ethics and professional behavior in nutrition

DTK3223 Kaunseling Pemakanan aan Pendidikan Pesakit *Nutrition/ Counseling and Patient Education* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi prinsip, teori kaunseling, komunikasi berkesan, dan kemahiran kaunseling dalam konteks amalan dietetik. Kaunseling individu dan kelompok dijalankan berdasarkan model proses penjagaan pemakanan. Teknik dalam pendidikan pesakit dititikberatkan. Kursus ini juga membincangkan etika dan tingkah laku profesional dalam kaunseling pemakanan dan pendidikan pesakit.

This course encompasses principles, counseling theory, effective communication, and counseling skills in the context of dietetic practice. Individual and group counseling are conducted based on the nutrition care process model. Techniques in patient education are emphasized. The course also discusses ethics and professional behavior in nutritionncounseling and patient education.

DTK3100 Prinsip Penyediaan Makanan/ *Principles of Food Preparation* 3 (2+1)

Prasyarat : PKK3200

Kursus ini merangkumi pengenalan kepada asas makanan, konsep asas persediaan makanan serta kesan perubahan fizikal, kimia dan nilai pemakanan pada makanan yang dimasak. Kursus ini turut menekankan prosedur makmal dalam mengaplikasi prinsip semasa persediaan makanan

This course includes introduction to the fundamentals of foods, concepts of basic food preparation as well as effect of physical, chemical and nutritional changes in cooked foods. This course also emphasizes laboratory procedures on applying the principles studied to the preparation of foods

DTK3106 Pengurusan Perkhidmatan Makanan Institusi I/ *Institutional Foodservice Management I* 3 (3+0)

Prasyarat : DTK3105

Kursus ini meliputi sistem perkhidmatan makanan dan prinsip pengurusan perkhidmatan makanan institusi. Komponen penting dalam sistem perkhidmatan makanan bermula dari perancangan menu kepada perolehan barangan ditekankan. Aspek kelestarian dan peluang aktiviti keusahawanan dalam perkhidmatan makanan turut didedahkan.

This course is covering food service systems and principles of institutional food service management. Important components of the food service system start from menu planning to procurement of goods are emphasized. Sustainability aspects and entrepreneurial activity opportunities in food service are also highlighted.)

DTK3107 Pengurusan Perkhidmatan Makanan Institusi II/ *Institutional Foodservice Management II* 3 (1+2)

Prasyarat : DTK3106

Kursus ini menekankan latihan amali pengeluaran makanan berkuantiti yang berkualiti. Pengawasan kos dan bahan dalam menjana pendapatan turut ditekankan. Sistem perkhidmatan makanan dan amalan lestari di pelbagai institusi turut didedahkan

This course emphasizes practical training in producing quantity and quality food. Cost and ingredient control in generating income are also emphasized. Food service systems and sustainable practices in various institutions are also exposed.)

DTK3210 Asas Imunologi Pemakanan/ *Basic Nutritional Immunology* 2 (2+0)

Prasyarat : Tiada

Kursus ini meliputi pengenalan kepada konsep imuniti dan sistem imun dengan memberi penekanan kepada komponen nutrient dalam pemeliharaan sistem imun yang normal. Mekanisme nutrien dalam induksi inflamasi, fisiologi sel imun dan penyakit berkaitan pemakanan turut diperincikan. Kesan pemakanan dan aktiviti fizikal terhadap sistem imun turut dibincangkan.

The course includes an introduction to the concept of immunity and the immune system with emphasis on nutrient components in the maintenance of normal immune system. Mechanisms of nutrients in the induction of inflammation, immune cell physiology and diet-related disease are also specified. Effects of diet and physical activity on the immune system are discussed

DTK3211 Patofisiologi dan Pengurusan Klinikal Penyakit/ *Pathophysiology and clinical management of diseases* 4 (4+0)

Prasyarat : Tiada

Kursus ini meliputi aspek patofisiologi penyakit yang berkaitan pemakanan. Ia juga merangkumi interpretasi aspek biokimia dan klinikal penyakit, disusuli dengan diagnosis, pengurusan perubatan dan farmakologi penyakit berkaitan pemakanan. Aplikasi dietetik dalam imunologi dan interaksi ubat-nutrien turut dibincangkan.

This course covers aspects of the pathophysiology of nutrition-related diseases. It also includes the interpretation of the biochemical and clinical aspects of these diseases, followed by diagnosis, medical management, and pharmacology of nutrition-related diseases. The application of dietetics in immunology and drug-nutrient interactions is also discussed.)

DTK3222	Penilaian Taraf Pemakanan aan Kesihatan Pesakit/ <i>Assessment of Nutritional and Health Status of Patients</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi penilaian pemakanan yang merupakan sebahagian daripada proses penjagaan pemakanan bagi menentukan taraf kesihatan pesakit. Pelbagai kaedah penilaian taraf pemakanan dan kesihatan pesakit turut ditekankan. Data penilaian pemakanan pesakit dibandingkan dengan piawai.

This course encompasses nutritional assessment, which is part of the nutritional care process to determine the health status of patients. Various methods of assessing nutritional status and patient health are also emphasized. Patient nutritional assessment data is compared with standards.)

DTK3231	Pengalaman Awal Klinikal Dietetik I/ <i>Early Dietetics Clinical Experience I</i>	1 (0+1)
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Prasyarat : Tiada

Kursus ini merangkumi pendedahan awal tentang sistem penjagaan kesihatan dan perkhidmatan dietetik. Pelajar didedahkan dengan perkhidmatan dietetik klinikal, perkhidmatan dietetik komuniti, pengurusan perkhidmatan makanan di institusi dan peranan pentadbiran di peringkat kementerian. Peranan ahli dietetik sebagai ahli profesional kesihatan turut dibincangkan.

This course covers an early exposure to healthcare systems and dietetic services. Students are introduced to clinical dietetic services, community dietetic services, food service management in institutions and administrative role in the ministry. The role of dietitians as healthcare professionals is also discussed.)

DTK3232	Pengalaman Awal Klinikal Dietetik III/ <i>Early Dietetics Clinical Experience II</i>	1 (0+1)
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Prasyarat : DTK3231

Kursus ini adalah kursus berasaskan pengalaman yang memberi pendedahan kepada perkhidmatan dietetik klinikal melalui aplikasi Proses Penjagaan Pemakanan. Penekanan diberikan kepada amalan tingkah laku profesional, etika dan adab dalam mendekati pesakit. Teknik kaunseling individu dan kelompok dibincangkan dalam perbincangan kumpulan

This course is an experience-based course that provides exposure to clinical dietetic services through the application of Nutrition Care Process. Emphasis is given to practice of professional behavior, ethics and manners in approaching patients. Individual and group counseling techniques are discussed in group discussion.

DTK3233	Pengalaman Awal Klinikal Dietetik III/ <i>Early Dietetics Clinical Experience III</i>	1 (0+1)
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Prasyarat : DTK3232

Kursus ini adalah kursus berasaskan pengalaman awal klinikal yang menekankan kemahiran dalam Proses Penjagaan Pemakanan kepada pesakit. Faktor sosio-budaya juga diambil kira dalam strategi intervensi dan pengawasan tabiat pemakanan pesakit. Dokumentasi penjagaan pesakit turut dibincangkan

This course is an early clinical experience-based course that emphasizes skills in the Nutrition Care Process for patients. Socio-cultural factors are also taken into account in the intervention strategies and monitoring of patients' eating habits Patient care documentation is also discussed.)

$3(2+1)$

Prasyarat : DTK3221

Kursus ini merupakan pengenalan kepada terapi pemakanan perubatan. Penekanan diberi kepada hubungkait antara penyakit dengan prinsip modifikasi diet terapeutik sebagai komponen penting dalam Proses Penjagaan Pemakanan bagi jagaan kesihatan primer. Kajian kes yang merangkumi perancangan, penyediaan dan penilaian diet terapeutik turut dijalankan.

This course is an introduction to medical nutrition therapy. It focuses on the relationship between diseases and the principles of therapeutic dietary modification as an essential component of Nutrition Care Process of primary health care. Case study that includes planning, preparation and evaluation of therapeutic diets is conducted.)

 $3(2+1)$

Prasyarat : DTK3224

Kursus ini memfokuskan penyakit yang lebih kompleks dengan menerapkan prinsip modifikasi diet terapeutik, mengambilkira status keterjaminan makanan pesakit dalam komponen pengawalan dan pengurusan penyakit. Kajian kes berdasarkan Proses Penjagaan Pemakanan turut ditekankan.

This course focuses on complex diseases that incorporate the principles of therapeutic diet modifications, considering patient's food security status as part of the control and management of the diseases. Case study based on the Nutrition Care Process is emphasized.

 $3(2+1)$

Prasyarat : DTK3225

Kursus ini memfokuskan kepada penjagaan pemakanan pesakit berkeperluan khusus dan kritikal dengan penerapan prinsip diet terapeutik termasuk pemberian sokongan pemakanan dalam komponen pengurusan penyakit. Kajian kes berdasarkan Proses Penjagaan Pemakanan turut ditekankan.

This course focuses on nutrition care of patients with special needs and critical incorporating the principles of therapeutic diets including nutrition support delivery. Case study based on the Nutrition Care Process is emphasized.

6 (0+6)

Prasyarat : DTK3107

Kursus ini merangkumi latihan dan pengalaman bekerja di pelbagai operasi perkhidmatan institusi makanan. Kemahiran teknikal dan berfikir secara kritis terhadap operasi perkhidmatan makanan dan proses penjanaan pendapatan secara profesional dan beretika juga ditekankan

This course encompasses training and work experience in various institutional food service operations. Technical skills and critical thinking on the operation of food services and income generation processes in a professional and ethical manner is also emphasized.)

DTK4912	Praktikum Profesional Dietetik Klinikal/ <i>Clinical Dietetic Professional Practicum</i>	14 (0+14)
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Prasyarat : DTK3226

Kursus ini meliputi praktikum profesional dietetik klinikal pesakit luar dan pesakit dalam wad. Ia membentuk kemahiran terapi pemakanan perubatan termasuk saringan, penilaian status pemakanan, diagnosis, perancangan, intervensi, pemantauan dan dokumentasi Proses Penjagaan Pemakanan pesakit. Kemahiran dietetik dan pengembangan profesional turut ditekankan.

This course includes out-patient and in-patient clinical dietetic professional practicum. It develops the Medical Nutrition Therapy skills including screening, assessment, diagnosis, planning, intervention, monitoring and documentation of the patients' Nutrition Care Process. Dietetic skills and professional development are also emphasized.

DTK4914	Praktikum Profesional Dietetik Elektif Klinikal/ <i>Elective Clinical Dietetic Professional Practicum</i>	4 (0+4)
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Prasyarat : Tiada

Kursus ini merupakan praktikum profesional dietetik elektif di mana pelajar akan mendalami peranan ahli dietetik di pelbagai institusi pilihan di dalam atau di luar negara. Ia merangkumi sintesis pengalaman mengenai skop amalan dietetik di pelbagai lapangan termasuklah bidang dietetik memuncul. Aspek etika dan profesionalisme yang spesifik kepada skop amalan dietetik tersebut turut ditekankan.

This course is an elective professional dietetic practicum where students will delve into the role of dietitians in various institutions within or outside of the country. It encompasses the synthesis of experiences regarding the scope of dietetic practice in various fields, including emerging areas in dietetics. Specific aspects of ethics and professionalism related to the scope of dietetic practice are also emphasized.

DTK4913	Praktikum Profesional Dietetik Komuniti/ <i>Community Dietetic Professional Practicum</i>	6 (0+6)
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Prasyarat : Tiada

Kursus ini meliputi praktikum profesional dietetik komuniti yang menekankan kepada penilaiar keperluan dan pengurusan program intervensi dietetik di peringkat komuniti. Kemahiran merancang, menjalankan dan menilai program intervensi dietetik komuniti turut ditekankan secara profesional.

This course covers the dietetic professional practicum which emphasizes the dietetic community needs assessment and program management of community dietetics interventions. Skills to design, conduct and evaluate community dietetic intervention program is also emphasized, professionally.

DTK4959	Disertasi Bachelor/ <i>Bachelor Dissertation</i>	6 (0+6)
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Prasyarat : Tiada

Kursus ini merangkumi perancangan penyelidikan, pengumpulan, analisis dan tafsiran data, penyediaan dan pembentangan tesis. Penekanan diberi kepada kaedah saintifik dan etika profesional dalam penyelidikan

This course encompasses planning of research, collection, analysis and interpretation of data, preparation and presentation of thesis. Emphasis is on scientific research approach and professional ethics in research

Jabatan Pemakanan / Department of Nutrition

PKK3001 Seminar: Isu semasa dalam Pemakanan dan Kesihatan/ Seminar:
Current Issues In Nutrition And Health 2 (1+1)

Prasyarat : Tiada

Kursus ini merangkumi isu dan perkembangan terkini dalam bidang pemakanan dan kesihatan. Penemuan terkini dalam bidang ini turut dibincangkan. Penekanan juga diberikan kepada penulisan artikel saintifik dan pembentangan seminar

The course covers issues and developments in the field of nutrition and health. The latest discoveries in this field are also discussed. Scientific articles writing and seminar presentations are emphasized.

PKK3002 Biostatistik dan Kaedah Penyelidikan Pemakanan/ *Biostatistics and
Research Methodology in Nutrition* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi asas biostatistik dalam penyelidikan saintifik dalam bidang pemakanan dan kesihatan komuniti. Reka bentuk dan perancangan penyelidikan, menganalisis dan menginterpretasi data, penulisan dan penyampaian hasil kajian dibincangkan

This course encompasses the basics of biostatistics in scientific research in the field of nutrition and community health. Research design and planning, data analysis and interpretation, report writing and presentation of research findings are discussed

PKK3003 Anatomi dan Fisiologi Manusia/ *Human Anatomy and Physiology* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi pengenalan kepada sistem fisiologi dan anatomi manusia. Fungsi dan hubungan antara sistem badan serta proses dalam badan dalam mengekalkan taraf kesihatan yang optimum dibincangkan

This course encompasses an introduction to human physiological and anatomical systems. Functions and relationships among the body's system as well as processes in the body in maintaining optimal health conditions are discussed

PKK3100 Makanan, Masyarakat dan Budaya/ *Food, Society and Culture* 3 (3+0)

Prasyarat : Tiada

Kursus ini meliputi perspektif evolusi sistem makanan manusia dan perkaitannya dengan die kesihatan dan penyakit. Ia juga menekankan tabiat dan corak diet serta maksud makanan dalam pelbagai budaya. Faktor penentu untuk tabiat dan corak diet dalam budaya yang berbeza dan implikasinya ke atas pemakanan dan kesihatan di peringkat individu, komuniti dan populasi dibincangkan

This course covers evolutionary perspectives of human food system and its relationship to diet, health and disease. It also emphasizes dietary habits and patterns and the meaning of foods in various cultures. Determinants of dietary habits and patterns in different cultures and their implications on nutrition and health at individual, community and population levels are discussed

PKK3104	Asas Pengurusan Perkhidmatan Makanan/ <i>Basic Foodservice Management</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi konsep pengurusan sistem perkhidmatan makanan dalam pengeluaran makanan yang berkuantiti dan berkualiti secara komersial dan bukan komersial. Komponen penting dalam pengeluaran makanan bermula daripada perancangan menu, pembelian, penyimpanan dan kawalan inventori ditekankan. Penjagaan kebersihan premis untuk memastikan keselamatan makanan dan kemahiran keusahawanan juga ditekankan

This course encompasses the concepts of foodservice systems management in producing quantity and quality of commercial and non-commercial foods. The important components in food production beginning from menu planning, purchasing, storage and inventory control are emphasized. Sanitary premise maintenance to ensure food safety and entrepreneurial skills are also emphasized

PKK3105	Komposisi dan Analisis Nutrien/ <i>Nutrient Composition and Analysis</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini membincangkan elemen penting dalam analisis nutrien dan kesesuaiannya dalam bidang sains pemakanan. Elemen penting dalam analisis nutrien termasuk persampelan, kawalan kualiti pembentangan data dan penggunaan data nutrien dalam pelbagai aplikasi pemakanan diberi penekanan. Penentuan kandungan makro- dan mikronutrien turut dilakukan

This course discusses important elements in nutrient analysis, and its suitability in the field of nutritional science. Important elements in nutrient analysis includes sampling, quality control, data presentation, and use of nutrient data in various nutrition applications are emphasised. The determination of macro- and micronutrient contents are carried out

PKK3107	Keselamatan Makanan untuk Kesihatan Pengguna/ <i>Food Safety for Consumer Health</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi aspek keselamatan makanan meliputi keracunan makanan, pengendalian pemprosesan dan penyimpanan makanan dalam perkhidmatan makanan. Analisis Bahaya Titik Kawalan (HACCP), aditif makanan, pelabelan pemakanan, peranan agensi kerajaan, industri makanan dan pengguna dalam memastikan makanan yang selamat turut dibincangkan

This course discusses food safety aspect includes food poisoning, handling, preparation, processing and storage in food services. Hazard Analysis Critical Control Point (HACCP), food additives and nutritional labelling, roles of government agencies, food industries and consumers to ensure food safety are discussed

PKK3108	Teknik Makmal dalam Penyelidikan Pemakanan/ <i>Laboratory Techniques in Nutritional Research</i>	2 (1+1)
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Prasyarat : PKK3105

Kursus ini membincangkan konsep dan teknik makmal dalam penyelidikan pemakanan. Pelajar akan menjalankan persampelan, penyimpanan, merekod, dan menganalisis sampel dan data. Prinsip dan aplikasi pelbagai teknik makmal dan instrumen juga akan dibincangkan

This course discusses the concepts and laboratory techniques in nutrition research. Students will carry out sampling, keeping record and sample and data analysis. The principles and application of various laboratory techniques and instrumentations are also discussed

PKK3200 Prinsip Pemakanan Manusia/ *Principles of Human Nutrition* 3 (3+0)

Prasyarat : Tiada

Kursus ini merangkumi prinsip asas pemakanan manusia. Masalah kesihatan yang berkaitan dengan nutrien dan kepentingan Saranan Pengambilan Nutrien, Pinggan Sihat dan Garis Panduan Pemakanan diberi penekanan

This course covers the basic principles of human nutrition. Health problems related to nutrients and the importance of Recommended Nutrient Intake (RNI), Healthy Plate and Malaysian Dietar Guidelines are emphasized

PKK3203 Pemakanan Dalam Kitaran Hidup/ *Nutrition Throughout the Life Cycle* 3 (3+0)

Prasyarat : PKK3200

Kursus ini merangkumi pertumbuhan dari peringkat persenyawaan sehingga dewasa serta pemakanan sebagai pengaruh persekitaran yang penting terhadap pertumbuhan dan perkembangan terutamanya pada tempoh kritikal. Implikasi malnutrisi dari segi biologi dan tingkahlaku terhadap pertumbuhan dan perkembangan dibincangkan

This course encompasses human growth from conception to adulthood as well as nutrition as an important environmental influence on growth and development especially during critical periods. Biological and behavioural implications of malnutrition on growth and development are discussed

PKK3204 Biokimia Pemakanan/ *Nutritional Biochemistry* 4 (3+1)

Prasyarat : PKK3200

Kursus ini merangkumi biokimia dan metabolisme nutrien dan perkaitannya di dalam memelihara homeostasis tubuh. Ketidakseimbangan nutrien dan kesan ke atas metabolisme tubuh, petunjuk biokimia di dalam gangguan pemakanan dan metabolisme diberi penekanan

This course encompasses biochemistry and metabolism of nutrients and their interactions in maintaining body homeostasis. Nutrient imbalance and its effect on body metabolism, biochemical indicators of nutritional disorders and inborn errors of metabolism are emphasized

PKK3206 Genetik dan Pemakanan/ *Genetics and Nutrition* 2 (2+0)

Prasyarat : Tiada

Kursus ini merangkumi pengetahuan asas dalam genetik, konsep asas gen dan komponennya; warisan dan pandangan mengenai teknologi genetik yang telah berkembang pada masa kini. Pelajar juga akan didedahkan dengan nutrigenomik dan interaksi nutrien dan gen. Implikasi terhadap kesihatan dan penyakit juga dibincangkan

This course covers fundamental knowledge in genetics, basic concept of genes and its components, inheritance and overview of genetic technologies which have been developing currently. Students also will be exposed to nutrigenomic and nutrient and gene interactions. Implications on health and diseases are discussed

PKK3207	Pemakanan Klinikal I/ <i>Clinical Nutrition I</i>	3 (3+0)
Prasyarat : Tiada		
Kursus ini menerangkan laluan biokimia dalam mendiagnosis penyakit berkaitan dengan pemakanan seperti diabetes, penyakit jantung, kanser, anemia dan prosedur standard yang perlu dipatuhi. Pelajar juga mempelajari konsep farmakologi, pengelasan drug, cara drug bertindak dan kesan sampingan drug. Data yang diperolehi dari ujian biokimia akan digunakan untuk memilih kaedah pengurusan penyakit yang bersesuaian		
<i>This course explains the biochemical pathway in diagnosis of nutrition-related diseases such as diabetes, heart disease, cancer, anaemia and standard procedure that need to be followed. Students also learn the concepts of pharmacology, drug classification, their action as well as the side effects. Data from biochemical assessment will be used to select the suitable management of the diseases</i>		
PKK3208	Pemakanan Klinikal I/ <i>Clinical Nutrition I</i>	3 (3+0)
Prasyarat : Tiada		
Kursus ini merangkumi aspek fisiologikal penyakit berkaitan dengan pemakanan dan hubungan antara penyakit dan nutrien yang terlibat. Prinsip dan konsep pemakanan terapi sebagai panduan dalam perancangan diet terapeutik dan kesan ubatan dalam merawat penyakit akan dibincangkan		
<i>This course encompasses physiological aspects of nutrition-related diseases and the relationship between disease and nutrient involved. Principles and concepts of nutritional therapy as guidelines in the planning of therapeutic diets and implications of drugs on disease treatment are discussed</i>		
PKK3302	Epidemiologi Asas/ <i>Basic Epidemiology</i>	3 (3+0)
Prasyarat : Tiada		
Kursus ini mengandungi prinsip asas epidemiologi dalam kesihatan komuniti dan klinikal, kaedah epidemiologi dalam penyelidikan dan kaedah statistik yang digunakan dalam epidemiologi		
<i>This course covers the basic principles of epidemiology in community and clinical health, epidemiologic methods in research and statistical methods used in epidemiology</i>		
PKK3303	Kesihatan Ibu dan Anak/ <i>Maternal and Child Health</i>	3 (2+1)
Prasyarat : PKK3203		
Kursus ini merangkumi kesihatan ibu dan anak termasuk kesalinggantungan antara ibu dan anak, pertumbuhan dan perkembangan yang optimum bagi kanak-kanak melalui pemakanan dan kesihatan. Keperluan perkhidmatan dan program di peringkat kebangsaan dan antarabangsa untuk meningkatkan status kesihatan ibu dan anak turut dibincangkan		
<i>This course encompasses maternal and child health which includes the interdependence between mothers and children, optimal growth and development of children through nutrition and health. The need for maternal and child health services and programmes at national and international levels to improve their health status are discussed</i>		
PKK3304	Tingkahtlaku dan Kesihatan/ <i>Behaviour and Health</i>	3 (3+0)

Prasyarat : Tiada

Kursus ini merangkumi pengetahuan berkaitan di antara tingkahlaku dan kesihatan. Ia juga memberi fokus terhadap faktor yang mempengaruhi, membolehkan dan memperkukuhkan tingkahlaku. Kaedah penyelidikan tingkahlaku dan kesihatan serta perkaitan di antara factor tingkahlaku dan status kesihatan juga ditekankan

This course encompasses knowledge on the relationship between behavior and health conditions. It will also focus on predisposing, enabling and reinforcing factors to behavior. Research methods in health and behavior and associations between behavioral factor and health status are emphasized

PKK3401 Penilaian Taraf Pemakanan dan Kesihatan/ *Assessment of Nutritional and Health Status* 3 (2+1)

Prasyarat : PKK3200

Kursus ini merangkumi penilaian taraf kesihatan dan pemakanan individu, kumpulan dan populasi. Ukuran antropometri, analisis sampel biologi, pemeriksaan klinikal, kaedah diet dan kajian pengaruh persekitaran, penilaian kuantitatif dan kualitatif turut ditekankan

This course encompasses the assessments of health and nutritional status of individuals, groups and populations. Anthropometric measurements, analysis of biological samples, clinical examination, dietary methods and study of environmental influences, the quantitative and qualitative assessments are emphasized

PKK3406 Promosi Pemakanan dan Kesihatan I/ *Nutrition and Health Promotion I* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi konsep, definisi dan perkembangan promosi kesihatan, serta proses perancangan, pelaksanaan dan penilaian program promosi pemakanan dan kesihatan

This course encompasses concept, definition and development of health promotion, as well as planning, implementation and evaluation processes of nutrition and health promotion programs

PKK3407 Promosi Pemakanan dan Kesihatan II/ *Nutrition and Health Promotion II* 2 (0+2)

Prasyarat : PKK3406

Kursus ini merangkumi konsep, prinsip dan fungsi utama pengurusan program promosi pemakanan dan kesihatan. Aktiviti perancangan, pelaksanaan dan penilaian program promosi pemakanan dan kesihatan di sekolah, komuniti dan tempat kerja akan dijalankan

This course encompasses concept, principle and main functions of management of nutrition and health program. Activities related to planning, implementation and evaluation of nutrition and health promotion programs in schools, communities and work sites will be conducted

PKK3408 Polisi Makanan dan Pemakanan/ *Food and Nutrition Policy* 3 (3+0)

Prasyarat : Tiada

Kursus ini merangkumi polisi makanan dan pemakanan di peringkat kebangsaan dan antarabangsa. Ia juga memberi penekanan kepada cabaran pemakanan global dalam pembentukan polisi makanan dan pemakanan. Polisi dan pelan tindakan yang berkaitan dengan makanan dan pemakanan peringkat kebangsaan turut dibincang

This course encompasses food and nutrition policies at national and international levels. It also emphasizes global nutrition challenges in the development of food and nutrition policy. Policy and plan of actions related to food and nutrition at national level are discussed

PKK4902 Latihan Industri/ *Industrial Training*

18 (0+18)

Prasyarat : Tiada

Kursus ini mengandungi pengetahuan mengenai masalah pemakanan dan kesihatan komuniti dan mengenal pasti cara untuk mengatasinya. Pelajar ditempatkan di agensi yang berkaitan dengan program pemakanan dan kesihatan komuniti serta diberi pengalaman bekerja agar pelajar dapat memahami dan boleh menjalankan aktiviti yang dirancang oleh agensi/organisasi berkenaan

This course encompasses knowledge on problems of the community and to identify means of overcoming them. Students will be placed in agencies related with nutritional and health community programs and be given work experiences so that students have an understanding and will be able to carry out activities that have been planned by the agency/organization

PKK4959 Disertasi Bachelo/ *Bachelor Dissertation*

12 (0+12)

Prasyarat : Tiada

Kursus ini merangkumi penyediaan cadangan penyelidikan, pelaksanaan dan penulisan saintifik untuk projek penyelidikan dalam bidang pemakanan dan kesihatan komuniti. Penekanan diberi kepada pelaporan dan pembentangan hasil penyelidikan

This course covers the preparation of research proposal, implementation and scientific writing of research project related to nutrition and community health. Emphasis is given on the reporting and presentation of research findings

PKK4202 Pemakanan Peringkat Remaja/ *Nutrition in Adolescence*

3 (3+0)

Prasyarat : PKK3203

Kursus ini merangkumi proses perkembangan remaja dan faktor yang mempengaruhi status pemakanan dan kesihatan. Penilaian taraf pemakanan serta strategi menambahbaik, mengekalkan dan menggalakkan taraf kesihatan remaja turut diberi penekanan

This course encompasses the developmental process and factors influencing the nutritional and health status of adolescents. The assessments of nutritional status as well as strategies to improve, maintain and promote the health status of adolescents are emphasized

PKK4205 Pemakanan, Aktiviti Fizikal dan Kesihatan/ *Nutrition, Physical Activities and Health*

3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi aspek aktiviti fizikal dan senaman, serta perkaitannya dengan pemakanan dan kesihatan. Penilaian tahap aktiviti fizikal bagi kumpulan yang terpilih akan dilakukan. Perancangan menu dan hubungan antara aktiviti sukan dan pemakanan juga akan ditekankan

This course encompasses the aspects of physical activity and exercise, and their relationship with nutrition and health. Assessment of physical activity levels of selected groups will be measured. Menu planning and the relationship between sport activities and nutrition will also be emphasized

PKK4206	Pemakanan Sukan/ <i>Sports Nutrition</i>	3 (3+0)
Prasyarat : PKK4205		
Kursus ini membincangkan konsep sukan, senaman dan kecergasan. Ia juga menerangkan peranan nutrien semasa dan selepas latihan. Hubungkait antara pemakanan, prestasi sukan dan pemulihan daripada latihan dan senaman akan diterangkan. Isu dan kontroversi yang berkaitan dengan pemakanan sukan seperti peranan makanan tambahan, hormon, minuman sukan, vitamin dan bantuan ergogenik akan turut dinilai. <i>This course discusses concept of sports, exercise and fitness. It describes the roles of nutrients during and after training. Interrelation between nutrition, athletic performance and recovery from exercise and training will be explained. Issues and controversies related to sport nutrition such as role of dietary supplements, hormonal supplements, sports drinks, vitamins and ergogenic aids will be critically evaluated</i>		
PKK4307	Kesihatan Pengguna/ <i>Consumer Health</i>	3 (3+0)
Prasyarat : Tiada		
Kursus ini merangkumi konsep dan isu kesihatan pengguna. Keputusan dan pemilihan barangan kesihatan pengguna, perkhidmatan dan kemudahan serta faktor yang mempengaruhi pemilihan turut dijelaskan. Undang-undang perlindungan pengguna yang berkaitan dengan barangan atau perkhidmatan kesihatan juga dibincangkan <i>This course encompasses the concept and issues of consumer health. The decisions and selections of consumer health products, services and facilities as well as factors affecting choices are explained. Consumer law and protection with respect to health products and services are also discussed</i>		
PKK4401	Kesihatan dan Pemakanan Geriatrik/ <i>Geriatric Health and Nutrition</i>	3 (3+0)
Prasyarat : Tiada		
Kursus ini merangkumi kesan proses penuaan pada perubahan fizikal dan fisiologi warga tua. Faktor yang mempengaruhi status pemakanan dan kaedah penilaian status pemakanan dan kesihatan warga tua dibincangkan. Masalah utama kesihatan dan pemakanan dalam kalangan warga tua yang berkaitan dengan proses penuaan turut ditekankan <i>This course covers the impact of aging process on physical and physiological changes of the elderly. Factors affecting nutritional status and methods of assessment of nutritional and health status of the elderly are discussed. The main age-related health and nutritional problems among the elderly are also emphasized</i>		
PKK4403	Praktikum Pemakanan Komuniti/ <i>Community Nutrition Practicum</i>	4 (0+4)
Prasyarat : Tiada		
Kursus ini merangkumi pelaksanaan aktiviti yang berkaitan dengan kesihatan untuk meningkatkan kesedaran dan kefahaman komuniti tentang kepentingan pemakanan dan kesihatan dalam kehidupan harian. Pengetahuan pemakanan dan kesihatan komuniti diaplikasikan melalui aktiviti kerjasama dengan komuniti <i>This course encompasses implementation of activities related to health to increase awareness and understanding of the community on the importance of nutrition and health in daily lives. Knowledge in nutrition and community health is applied through joined activities with the community</i>		

Jabatan Perubatan/ Department of Medicine

MDR3001 Biologi Manusia/ *Human Biology*

3

Prasyarat : Tiada

Kursus ini merangkumi organisasi, fungsi asas sel, tisu dan organ serta komunikasi sel. Ia juga merangkumi aspek asas anatomi kasar otot, tulang dan sistem saraf pusat dan periferi serta perkembangan awal janin

The course encompasses the organization, basic functions of cells, tissues and organs, as well as cellular communication. It also encompasses the gross anatomy of muscle, bone, central and peripheral nervous system as well as general embryology

MDR3002 Biokimia dan Genetik Manusia/ *Human Biochemistry and Genetics*

3

Prasyarat : Tiada

Kursus ini merangkumi struktur dan fungsi biomolekul, konsep asas genetik, metabolisme dan pengawalaturannya. Pelajar turut didedahkan kepada aplikasi genetik dan biokimia dalam bidang perubatan

This course encompasses the structure and function of biomolecules, basic genetic, metabolism and regulation of biomolecules. It also introduces students on the application of genetic and biochemistry in medicine

MDR3003 Pengenalan kepada Patologi Am/ *Introduction To General Pathology*

2

Prasyarat : Tiada

Kursus ini merangkumi prinsip asas patologi yang merangkumi patologi am dan patogen serta hubungkait dengan pengetahuan anatomi, fisiologi dan biokimia. Ia merangkumi aspek etiologi, patogenesis, perubahan morfologi dan kepentingan klinikal

The course encompasses principles of basic pathology which include general pathology and pathogens and relates with anatomy, physiology and biochemistry. It covers aspects of aetiology, pathogenesis, morphological changes and clinical significance

MDR3004 Farmakologi Am dan Biokimia/ *General and Biochemical Pharmacology*

2

Prasyarat : Tiada

Kursus ini merangkumi konsep farmakodinamik dan farmakokinetik termasuk pelbagai kesan klinikal, hubungkait dos-tindak balas, serta interaksi di antara ubat-ubatan. Ubat-ubatan kemoterapi dan ubat sistem saraf autonomi turut diperkenalkan

This course encompasses concepts on pharmacodynamics and pharmacokinetics including various types of clinical drug effects, dose-response relationship, and possible drugs interactions. It also introduces chemotherapeutic and autonomic drugs

MDR3005 Sistem Muskuloskeletal/ *Musculoskeletal System*

4

Prasyarat : Tiada

Kursus ini merangkumi pengenalan kepada anatomi, histologi dan fisiologi sistem muskuloskeletal. Patologi dan farmakologi sistem tersebut turut dibincangkan

The course encompasses the introduction to anatomy, histology and physiology of the musculoskeletal system. It also introduces students on the pathology and pharmacology of the system

MDR3006	Sistem Kardiovaskular/ <i>Cardiovascular System</i>	3
Prasyarat : Tiada		
Kursus ini merangkumi bidang anatomi, fisiologi, patologi dan farmakologi sistem kardiovaskular. Kursus ini turut menekankan manifestasi klinikal penyakit kardiovaskular, prinsip rawatan dan langkah pencegahan penyakit		
<i>This course encompasses the disciplines of anatomy, physiology, pathology and pharmacology of the cardiovascular system. This course also emphasizes on clinical manifestations of cardiovascular diseases, principles of treatment and preventive measures of diseases</i>		
MDR3007	Sistem Respiratori/ <i>Respiratory System</i>	3
Prasyarat : Tiada		
Kursus ini merangkumi struktur anatomi dan fungsi sistem respiratori, serta perubahan patologi yang berkaitan dalam hipoksia, kegagalan respiratori dan persekitaran luar biasa. Ini juga termasuk manifestasi klinikal, diagnosis dan rawatan penyakit berkaitan sistem respiratori		
<i>This course encompasses the anatomical structures and functions of the respiratory system, as well as the related pathological changes in hypoxia, respiratory failure and unusual environment. It also includes clinical manifestations, diagnosis and treatment of respiratory system-related diseases</i>		
MDR3008	Sistem Alimentari / <i>Alimentary System</i>	3
Prasyarat : Tiada		
Kursus ini merangkumi pengenalan kepada anatomi, histologi dan fisiologi sistem alimentari. Aspek nutrisi, patologi dan farmakologi sistem tersebut turut dibincangkan		
<i>The course encompasses the introduction to anatomy, histology and physiology of the alimentary system. It also introduces students on the nutritional aspect, pathology and pharmacology of the system</i>		
MDR3009	Sistem Urinari/ <i>Urinary System</i>	3
Prasyarat : Tiada		
Kursus ini merangkumi pengenalan kepada anatomi, histologi dan fisiologi sistem urinari. Ia juga merangkumi mekanisme hormon yang mempengaruhi fungsi ginjal, peranan ginjal dalam pengawalan tekanan darah dan pengekalan keseimbangan asid-bes. Patologi dan farmakologi sistem tersebut turut dibincangkan		
<i>The course encompasses the introduction to anatomy, histology and physiology of the urinary system. It also covers the hormonal mechanisms which affect renal functions, the role of kidneys in the control of blood pressure and maintenance of acid-base balance. It also introduces students on the pathology and pharmacology of the system</i>		
MDR3010	Sistem Reproduksi/ <i>Reproductive System</i>	3
Prasyarat : Tiada		
Kursus ini merangkumi pengenalan kepada anatomi, histologi, fisiologi, patologi klinikal dan farmakologi di dalam sistem reproduktif lelaki dan wanita.		
<i>The course encompasses introduction of anatomy, histology, physiology, clinical pathology and pharmacological of the male and female reproductive systems</i>		

MDR3011	Sistem Endokrin/ <i>Endocrine System</i>	3
Prasyarat : Tiada		
Kursus ini merangkumi bidang anatomi, histologi, fisiologi, patologi dan farmakologi yang berkaitan dengan sistem endokrin. Aspek normal dan abnormal sistem endokrin yang melibatkan rembesan hormon, paksi hormon dan pengawalan pengeluaran hormon turut dibincangkan		
<i>The course encompasses the disciplines of anatomy, histology, physiology, pathology and pharmacology related to the endocrine system. The normal and abnormal aspects of the endocrine system, which involves hormone secretion, hormonal axis, and regulation of hormone release are also discussed</i>		
MDR3012	Sistem Saraf Pusat/ <i>Central Nervous System</i>	4
Prasyarat : Tiada		
Kursus ini merangkumi anatomi, fisiologi dan patologi otak serta saraf tunjang. Pengetahuan ubatan untuk rawatan sistem saraf pusat yang lazim, termasuk mekanisma tindakbalas, efikasi dan had klinikal turut dibincangkan		
<i>This course encompasses the anatomy, physiology and pathology of the brain and spinal cord. The knowledge of drugs for treatment of common central nervous system disorders, including the mechanisms of action, clinical efficacy and limitations is also been discussed</i>		
MDR3013	Hematologi dan Imunologi / <i>Haematology and Immunology</i>	4
Prasyarat : Tiada		
Kursus ini merangkumi perubahan yang berlaku kepada organisasi, fungsi asas, morfologi sel darah dan komponennya dalam penyakit hematologi lazim. Konsep imuniti, tindak balas imun dalam gangguan sistemik dan ujian makmal yang berkaitan turut dibincangkan		
<i>The course encompasses the changes which occur to the organisation, basic functions, morphology of blood cells and their components in common haematological disorders. Concept of immunity, immune response towards systemic disorders and relevant laboratory investigations will be discussed</i>		
MDR3014	Penyakit Berjangkit/ <i>Infectious Disease</i>	3
Prasyarat : Tiada		
Kursus ini merangkumi pengetahuan tentang mikroorganisma penyebab penyakit berjangkit. Ia juga meliputi ciri, cara penyebaran, patogenesis dan manifestasi klinikal penyakit tersebut. Diagnosis dan prinsip rawatan dan pencegahan penyakit berjangkit juga ditekankan		
<i>This course encompasses the knowledge of causative microorganisms in infectious diseases. It also covers the characteristics, modes of transmission, pathogenesis and clinical manifestations of the disease. Diagnosis and principle of management of infectious diseases are emphasised</i>		
MDR3101	Profesionalisme dan Sahsiah I/ <i>Professionalism and Personal Development I</i>	2
Prasyarat : Tiada		
Kursus ini memperkenalkan ciri-ciri profesionalisme perubatan, prinsip asas etika perubatan, kemahiran komunikasi dalam hubungan doktor dan pesakit. Prinsip perubatan berasaskan bukti, amalan refleksi, kompetensi budaya serta resilien diperkenalkan		
<i>This course introduces the attributes of medical professionalism, basic principles of medical ethics, communication skills and doctor-patient relationships. Principles of evidence-based medicine, reflective practices, cultural competency as well as resilience will be introduced</i>		

MDR3102	Profesionalisme dan Sahsia II/ <i>Professionalism and Personal Development II</i>	3
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Prasyarat : Tiada

Kursus ini menekankan dengan lebih mendalam peranan, sikap dan kemahiran profesional seorang doktor perubatan dalam hubungan doktor-pesakit yang kompleks dan juga sesama petugas kesihatan. Kemahiran komunikasi lanjutan seperti penyampaian berita buruk dan pengendalian pesakit yang sukar akan diperkenalkan. Aspek resilien juga turut diberikan penekanan

This course further emphasizes the role, attitude and professional practices of a doctor in more complex doctor-patient relationships as well as interprofessional collaborative practice. Advanced communication skills such as breaking bad news and handling difficult patient encounters are introduced in this course. Resilience as thriving physicians are also emphasized

MDR3103	Pengalaman Awal Klinikal/ <i>Early Clinical Experience</i>	2
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Prasyarat : Tiada

Kursus ini memberi pendedahan kepada pengambilan sejarah perubatan dan pemeriksaan fizikal pesakit. Ia juga melatih kemahiran komunikasi berkesan dan kaedah prosedur asas klinikal

This course exposes the students to patient history taking and physical examination. It also involves effective communication skills and basic clinical procedures

MDR3104	Pengalaman Kemanusiaan dan Kepimpinan/ <i>Humanitarian and Leadership Experience</i>	1
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Prasyarat : Tiada

Kursus ini melibatkan aspek kesukarelawanan melalui perancangan dan pelaksanaan khidmat komuniti. Kursus ini juga turut menekankan kemahiran kepemimpinan dan kepentingan pembelajaran sepanjang hayat

This course involves volunteerism through planning and implementation of community services. It also emphasizes on leadership skills and importance of lifelong learning

MDR3105	Peperiksaan Profesional I/ <i>Professional Examination I</i>	3
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Prasyarat : Lulus semua kursus Tahun 1 dan Tahun 2

Peperiksaan ini menilai aplikasi pengetahuan sains perubatan asas kesemua sistem badan manusia. Peperiksaan ini turut menilai kemahiran asas klinikal dan komunikasi berkesan

This examination evaluates application of medical sciences knowledge of the human body system. It also assesses basic clinical skills and effective communications

MDR3106	Peperiksaan Profesional II/ <i>Professional Examination II</i>	3
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Prasyarat : Tiada

Peperiksaan ini menilai aspek komprehensif mengenai pengambilan sejarah pesakit dan pemeriksaan fizikal, pendekatan yang rasional dalam penyelesaian dan pengendalian masalah klinikal, aspek terapeutik bidang pereubatan dan prosedur rawatan

This examination evaluates comprehensive aspects of history taking and physical examination, rational approach on investigation and management of clinical problems, therapeutic aspects of medical practice and therapeutic procedures

MDR3107	Keselamatan Pesakit/ <i>Patient Safety</i>	1
Prasyarat : Tiada		
Kursus ini memperkenalkan prinsip keselamatan pesakit termasuk kesilapan lazim sewaktu penjagaan pesakit serta langkah pencegahannya. Ia juga mendedahkan peranan doktor perubatan dalam penambahbaikan keselamatan pesakit		
<i>This course introduces the principles of patient safety including common errors and their preventions. It also exposed the roles of medical professionals in improving patient safety</i>		
MDR3201	Perubatan Komuniti Asas/ <i>Fundamental Community Medicine</i>	2
Prasyarat : Tiada		
Kursus ini merangkumi konsep asas, prinsip dan isu berkaitan dengan perubatan kesihatan komuniti iaitu meliputi perubatan kesihatan keluarga, perubatan kesihatan pekerjaan, perubatan kesihatan persekitaran serta pendekatan menyeluruh melalui promosi kesihatan dan sistem penjagaan kesihatan. Faktor risiko penyakit, pencegahan dan kawalan dengan menggunakan kaedah epidemiologi turut ditekankan		
<i>This course encompasses basic concepts, principles and issues related to community health medicine including family health medicine, occupational health medicine, environmental health medicine and holistic approach through health promotion and health care system. It also emphasises disease risk factors, prevention and control, using epidemiological methods</i>		
MDR3202	Perubatan Komuniti/ <i>Community Medicine</i>	5
Prasyarat : Tiada		
Kursus ini merangkumi amalan dan aktiviti kesihatan awam di peringkat klinik kesihatan, pejabat kesihatan daerah dan hospital daerah dari segi kawalan pencegahan penyakit, promosi kesihatan dan penjagaan pesakit. Ia juga meliputi penilaian taraf kesihatan komuniti dan pengendalian program intervensi kesihatan		
<i>The course encompasses the practices and activities of public health activities in health clinics, District Health Office and District Hospital which include disease prevention, health promotion and patient care. Evaluation of the community health status and implementation of health intervention programme will also be covered in this course</i>		
MDR3203	Perubatan Pengenalan/ <i>Introductory Medicine</i>	11
Prasyarat : Tiada		
Kursus ini meliputi latihan dalam aspek asas pengambilan sejarah perubatan dan sejarah pesakit, pemeriksaan fizikal, pendekatan yang bersesuaian dalam mengendalikan masalah klinikal, aspek terapeutik bidang perubatan dan prosedur rawatan di dalam wad.		
<i>The course encompasses basic aspects of history taking and physical examination, appropriate approach on investigation and management of clinical problems, therapeutic aspects of medical practice and therapeutic procedures in the wards</i>		
MDR3204	Perubatan Lanjutan/ <i>Senior Medicine</i>	7
Prasyarat : Tiada		
Kursus ini meliputi latihan dalam aspek yang komprehensif mengenai pengambilan sejarah pesakit dan pemeriksaan fizikal, pendekatan yang rasional dalam penyelesaian dan pengendalian masalah klinikal, aspek terapeutik bidang perubatan dan prosedur rawatan di dalam wad		
<i>The course encompasses comprehensive aspects of history taking and physical examination, rational approach on investigation and management of clinical problems, therapeutic aspects of medical practice and therapeutic procedures in the wards</i>		

MDR3205	Surgeri Pengenalan/ <i>Introductory Surgery</i>	11
Prasyarat : Tiada		
Kursus ini meliputi latihan dalam aspek asas pengambilan sejarah pesakit dan pemeriksaan fizikal, pendekatan yang bersesuaian dalam mengendalikan masalah klinikal, aspek terapeutik dan prosedur rawatan di dalam bidang surgeri. Pelajar juga didedahkan kepada rawatan pesakit trauma, mengalami renjatan dan mangsa kebakaran		
<i>The course encompasses basic aspects of history taking and physical examination, appropriate approach on investigation and management of clinical problems, therapeutic aspects of medical practice and therapeutic procedures in surgery. It also introduces the students to trauma, shock and burn management</i>		
MDR3206	Surgeri Lanjutan/ <i>Senior Surgery</i>	7
Prasyarat : Tiada		
Kursus ini menekankan dengan lebih mendalam aspek pengurusan klinikal dalam bidang surgeri. Pelajar turut didedahkan dengan kaedah rasional dalam strategi dan penyiasatan penyakit berdasarkan diagnosis surgeri yang berkaitan dengan peribedah, rawatan pesakit luar serta prosedur rawatan di pelbagai fasiliti surgeri		
<i>This course provides greater emphasis on clinical management in surgery. It also covers rational approach on strategy and investigations based on the working and differential diagnoses leading to perioperative, outpatient management and therapy of surgical patients in various surgical facilities</i>		
MDR3207	Perubatan Keluarga Pengenalan/ <i>Introductory Family Medicine</i>	4
Prasyarat : Tiada		
Kursus ini memperkenalkan amalan serta spektrum penyakit dan pesakit di peringkat rawatan primer serta prinsip perubatan keluarga agar pelajar dapat menghubungkan proses manifestasi klinikal pesakit dan penyelesaian pelbagai masalah klinikal di peringkat primer		
<i>This course introduces a practice of primary care medicine, the spectrum of diseases and patients seen in primary care setting and principles of family medicine to enable the students to relate the process of patients' clinical manifestations and problem solving in multiple clinical situations in primary care</i>		
MDR3208	Perubatan Keluarga Lanjutan/ <i>Senior Family Medicine</i>	4
Prasyarat : Tiada		
Kursus ini memantapkan amalan perubatan keluarga di peringkat kesihatan primer agar pelajar dapat menghubungkan proses yang terlibat dalam manifestasi klinikal pesakit dengan menyelesaikan pelbagai masalah klinikal		
<i>This course strengthens the practice of family medicine at the primary care level. This enables the students to relate the process of patients' clinical manifestation and problem solving in multiple clinical situations</i>		

MDR3209	Penjagaan Kesihatan Warga Tua/ <i>Healthcare in Elderly</i>	3
Prasyarat : Tiada		
Kursus ini merangkumi demografi penuaan, penilaian pesakit, prinsip perubatan, masalah psikologi, masalah kejatuhan, imobiliti, rehabilitasi, nazak dan kematian di kalangan pesakit warga tua. Etika dan profesionalisme juga di berikan penekanan. Kepentingan penuaan yang sihat diperkukuhkan dalam program ini		
<i>This course encompasses aging demography, assessment of elderly patients, principles of geriatric medicine, psychological problems in the elderly, falls, immobility, rehabilitation, dying and death. Ethics and professionalism in managing the elderly is also highlighted. The importance of healthy and graceful ageing is incorporated in the program</i>		
MDR3210	Ortopedik Pengenalan/ <i>Introductory Orthopaedic</i>	3
Prasyarat : Tiada		
Kursus ini meliputi latihan dalam aspek asas pengambilan sejarah dan pemeriksaan fizikal pesakit ortopedik, pendekatan yang bersesuaian dalam mengendalikan masalah klinikal, aspek terapeutik bidang ortopedik dan prosedur rawatan di dalam wad		
<i>The course encompasses basic aspects of history taking and physical examination of orthopaedic patients, appropriate approach on investigation and management of clinical problems, therapeutic aspects of orthopaedic practice and therapeutic procedures in the wards</i>		
MDR3211	Ortopedik Lanjutan/ <i>Senior Orthopaedic</i>	6
Prasyarat : Tiada		
Kursus ini meliputi latihan dalam aspek pengambilan sejarah dan pemeriksaan fizikal pesakit ortopedik, pendekatan yang bersesuaian dalam mengendalikan masalah klinikal, aspek terapeutik bidang ortopedik dan prosedur rawatan di dalam wad, unit kecemasan, unit fisioterapi, unit jurupulih carakerja, unit ortosis dan prosthesis		
<i>The course encompasses aspects of history taking and physical examination of orthopaedic patients, appropriate approach on investigation and management of clinical problems, therapeutic aspects of orthopaedic practice and therapeutic procedures in the wards, emergency unit, physiotherapy, occupational therapy, orthotic and prosthesis units</i>		
MDR3212	Radiologi Pengenalan/ <i>Introductory Radiology</i>	1
Prasyarat : Tiada		
Kursus ini merangkumi asas modaliti pengimejan yang berbeza, termasuk bahaya sinaran dan perlindungan sinaran, aspek prosedur pemeriksaan, indikasi klinikal serta kontraindikasi dan pentafsiran asas imej dan modaliti radiologi untuk penyakit lazim		
<i>The course encompasses the basics in different imaging modalities which include radiation hazards and protection, the aspect of examination procedures, their clinical indications and contraindications and also covers basic interpretation of radiological modalities and images of common diseases</i>		
MDR3213	Radiologi Klinikal/ <i>Clinical Radiology</i>	1
Prasyarat : Tiada		
Kursus ini meliputi pentafsiran imej dan modaliti radiologi penyakit lazim. Kesesuaian pemeriksaan radiologi yang diperlukan turut ditekankan		
<i>The course covers the interpretation of radiological modalities and images of common diseases. Indication of each radiological examination needed is emphasized</i>		

MDR3214	Sokongan Hayat Asas/ <i>Basic Life Support</i>	1
Prasyarat : Tiada		
Kursus ini merangkumi aspek asas pemulihan kardiopulmonari untuk dewasa dan kanak-kanak. Ia juga merangkumi teknik rawatan kepada mangsa-mangsa tercekik. Pelajar juga mempelajari ubat dan peralatan yang digunakan untuk resusitasi di hospital		
<i>This course encompasses basic aspects of cardiopulmonary resuscitation in adults and children. It also include approach and technique during choking. Students will also learn the use of medications and equipments for resuscitation in hospital setting</i>		
MDR3215	Farmakologi Klinikal/ <i>Clinical Pharmacology</i>	3
Prasyarat : Tiada		
Kursus ini merangkumi aplikasi prinsip farmakologi dan penggunaannya mengikut garis panduan klinikal. Kursus ini juga menekankan perancangan rawatan penyakit dengan berpanduan indikasi keberkesanan dan kesan sampingan ubat-ubatan		
<i>This course covers the application of pharmacological principles according to the clinical guidelines. This course also emphasises on management plan based on indication, effectiveness and side effects of drugs</i>		
MDR3216	Patologi Klinikal/ <i>Clinical Pathology</i>	3
Prasyarat : Tiada		
Kursus ini merangkumi aplikasi prinsip asas penyiasatan patologi klinikal dalam rawatan pesakit. Ia juga meliputi indikasi isu berkaitan keselamatan pesakit dan pengamal perubatan profesional		
<i>This course covers the application of laboratory investigation principles in patient management. It also encompasses issues related to patient and healthcare professional safety</i>		
MDR3217	Obstetrik dan Ginekologi Pengenalan/ <i>Introductory Obstetrics and Gynaecology</i>	9
Prasyarat : Tiada		
Kursus ini merangkumi pengambilan sejarah pesakit dan pemeriksaan fizikal pada pesakit obstetrik dan ginekologi. Pelajar turut didedahkan dengan penyiasatan dan pengendalian masalah klinikal, aspek terapeutik serta prosedur rawatan di klinik, wad, bilik bersalin dan dewan bedah		
<i>The course encompasses history taking and physical examination of obstetrics and gynaecology patients. Students are exposed to investigation and management of clinical problems, therapeutic aspects and treatment procedures in the clinics, wards, labour rooms and operation theatres</i>		
MDR3218	Obstetrik dan Ginekologi Lanjutan/ <i>Senior Obstetrics and Gynaecology</i>	4
Prasyarat : Tiada		
Kursus ini menekankan dengan lebih mendalam aspek pengurusan klinikal dalam bidang obstetrik dan ginekologi. Pelajar turut didedahkan dengan kaedah rasional dalam strategi dan penyiasatan penyakit berdasarkan diagnosis obstetrik dan ginekologi yang berkaitan dengan peribedah, rawatan pesakit luar serta prosedur rawatan di pelbagai fasiliti obstetrik dan ginekologi		
<i>This course provides greater emphasis on clinical management in obstetrics and gynaecology. It also covers rational approach on strategy and investigations based on the working and differential</i>		

diagnoses leading to perioperative, outpatient management and therapy of patients in various obstetrics and gynaecology facilities

MDR3219 Pediatrik Pengenalan/ *Introductory Paediatrics* 9

Prasyarat : Tiada

Kursus ini merangkumi pengenalan kepada pediatrik am, kemahiran asas pengambilan sejarah perubatan dan teknik asas pemeriksaan fizikal pesakit kanak-kanak

This course encompasses the introduction to general paediatrics, basic skills of history taking and basic examination techniques in paediatric patients

MDR3220 Pediatrik Lanjutan/ *Senior Paediatrics* 4

Prasyarat : Tiada

Kursus ini merangkumi pengetahuan mengenai pediatrik am, pendekatan dalam pencegahan serta penyelesaian dan pengendalian masalah kesihatan yang lazim ditemui di kalangan kanak-kanak

This course encompasses knowledge in general paediatrics and approach to prevention and management of common health problems found in children

MDR3221 Psikiatri/ *Psychiatry* 7

Prasyarat : Tiada

Kursus ini merangkumi semua aspek klinikal penyakit psikiatri. Penekanan akan dibuat pada teknik temuduga, pengambilan sejarah perubatan pesakit, dan pemeriksaan status mental. Kebolehan mengenal pasti simptom, faktor yang menyumbang kepada penyakit mental serta pengurusan penyakit psikiatri lazim juga dititik beratkan

This course covers all the clinical aspects of psychiatric disorders. Emphasis will be made on interview techniques, history taking, and mental state examination. Recognition of symptoms of major psychiatric disorders as well as the understanding of various factors contributing to mental illness, case formulation and patient management are also highlighted

MDR3222 Otorinolaringologi/ *Otorhinolaryngology* 3

Prasyarat : Tiada

Kursus ini meliputi latihan dalam aspek yang komprehensif mengenai pengambilan sejarah pesakit dan pemeriksaan fizikal, pendekatan yang rasional dalam penyelesaian dan pengendalian masalah klinikal, aspek terapeutik bidang perubatan dan prosedur rawatan di dalam wad otorinolaringologi

The course encompasses comprehensive aspects of history taking and physical examination, rational approach on investigation and management of clinical problems, therapeutic aspects of medical practice and therapeutic procedures in the otorhinolaryngology wards

MDR3223	Oftalmologi/ <i>Ophthalmology</i>	3
Prasyarat : Tiada		
Kursus ini merangkumi ilmu oftalmologi asas di mana meliputi pengambilan sejarah pesakit, pemeriksaan fizikal dan pengurusan masalah mata yang lazim ditemui, termasuk yang disebabkan oleh jangkitan kuman, kecederaan, katarak, glaukoma dan keabnormalan retina. Pelajar dilatih untuk dapat menilai dan menangani kes-kes mata lazim dan merawat secara profesional		
<i>This course covers the basic ophthalmologic knowledge, which includes the collection of patient history, physical examination and common eye management problems, including those caused by infections of germs, injuries, cataracts, glaucoma and retina abnormalities. Students are trained to assess and handle the common eye cases and treat them professionally</i>		
MDR3224	Pengalaman Klinikal Luaran/ <i>External Clinical Experience</i>	3
Prasyarat : Tiada		
Kursus ini melibatkan latihan sangkut dalam bidang perubatan dan kesihatan yang diminati di fasiliti kesihatan luar untuk mendapatkan pengalaman bagi pembangunan profesionalisme dan sahsiah diri		
<i>This course encompasses attachment to medical and health disciplines of interest in external health facilities to gain experience for personal and professional development</i>		
MDR3225	Anestesiologi Klinikal/ <i>Clinical Anaesthesiology</i>	2
Prasyarat : Tiada		
Kursus ini merangkumi asas anestesiologi klinikal termasuk persediaan pra-pembiusan, pengurusan pesakit semasa surgery, penjagaan pesakit kritikal dan pengurusan kesakitan di dewan bedah dan di unit rawatan rapi dengan menitikberatkan peranan etika perubatan dan profesionalisme		
<i>This course encompasses basic clinical anaesthesiology inclusive of pre-anaesthetic preparation, peri-operative, critical care and pain management in operation theatre and intensive care unit with attention to ethics and professionalism</i>		
MDR3226	Perubatan Kecemasan/ <i>Emergency Medicine</i>	2
Prasyarat : Tiada		
Kursus ini merangkumi prinsip pengurusan akut kes kecemasan dalam menstabilkan dan mencegah kemerosotan keadaan pesakit. Ia menekankan kepentingan komunikasi dan kerjasama yang baik dengan unit klinikal lain bagi kesinambungan pengurusan pesakit. Kursus ini juga memberi pendedahan kepada penjagaan pra-hospital dan pengurusan bencana		
<i>This course encompasses the principles of acute management of emergency cases, which aim to stabilise and prevent the deterioration of patients' conditions. It emphasizes the importance of good communication and teamwork with other clinical units for the continuity of patient management. This course also give exposure to the pre-hospital care and disaster management</i>		

MDR3227	Perubatan Forensik/ <i>Forensic Medicine</i>	1
Prasyarat : Tiada		
Kursus ini merangkumi pengetahuan asas bidang forensik dan garis panduan dalam melakukan bedah siasat kes lazim seperti kematian mengejut, kemalangan, asfiksia dan kes medikolegal serta menulis laporan bedah siasat		
<i>This course encompasses basic knowledge of forensic and guidelines to perform autopsy on common cases such as sudden death, accidents, asphyxia and well as medicolegal cases and preparing post-mortem reports</i>		
MDR3901	Projek Penyelidikan/ <i>Research Project</i>	6
Prasyarat : Tiada		
Kursus ini merangkumi konsep asas projek dan kaedah penyelidikan, termasuk penggunaan statistik perubatan, rekabentuk penyelidikan dan penyediaan kertas cadangan penyelidikan. Kursus ini juga melibatkan pengumpulan dan analisis data, penulisan saintifik dan pembentangan projek.		
<i>This course encompasses the basic concepts of research methodology and project, including the use of medical statistics, research designs and proposal preparation. It also involves data collection and analysis, scientific write-up and project presentation</i>		

Jabatan Kejururawatan dan Rehabilitasi/ Department of Nursing and Rehabilitation

NUR3002 Pendidikan dalam Kejururawatan/ *Teaching in Nursing*

2(1+1)

Prasyarat : Tiada

Kursus ini meliputi prinsip dan strategi pengajaran dan pembelajaran dalam bidang kejururawatan. Kemahiran pengajaran dan pembelajaran ditekankan. Kaedah penilaian pengajaran yang berkesan didedahkan

This course covers the principles and strategies of teaching and learning in nursing. Teaching and learning skills are emphasized. Assessment methods of effective teaching are exposed

NUR3003 Perspektif Teori Kejururawatan/ *Perspective of Nursing Theories*

2(2+0)

Prasyarat : Tiada

Kursus ini merangkumi teori dan proses kejururawatan dalam amalan kejururawatan profesional. Hubungan kait antara teori kejururawatan dengan proses jagarawatan ditekankan

This course encompasses theories and nursing process in professional nursing practice. The relationship between nursing theories and nursing process are emphasized

NUR3004

4(3+1)

Prasyarat : Tiada

Kursus ini merangkumi konsep, kaedah, dan pelaksanaan penyelidikan secara kuantitatif, kualitatif dan epidemiologi dalam penyelidikan saintifik bidang kejururawatan. Kepentingan etika, pencarian literatur dan penulisan saintifik ditekankan

This course encompasses the concepts, methodology, and implementation of quantitative, qualitative research and epidemiology in nursing scientific resaerch. The importance of ethics, literature search and scientific writing are emphasized.

NUR3104 Biokimia dan Farmakologi/ *Biochemistry and Pharmacology*

3(2+1)

Prasyarat : Tiada

Kursus ini meliputi biokimia dan farmakologi dalam bidang kejururawatan. Penggunaan pengetahuan biokimia dalam menginterpretasi laporan makmal diaplikasikan. Pengetahuan farmakologi berkaitan dengan penggunaan ubatan secara selamat kepada klien ditekankan. Tatacara pemberian ubat berdasarkan prinsip pemberian ubat difokuskan

This course covers biochemistry and pharmacology in nursing. Knowledge of biochemistry is applied in interpreting laboratory report. Knowledge of pharmacology related to the safe usage of drugs are emphasized. Procedure of serving medication according to the principles are focused

NUR3105 Anatomi dan Fisiologi I/ *Anatomy dan Physiology I*

3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi konsep asas anatomi dan fisiologi manusia. Sistem tisu asas badan, integumen, muskuloskeletal, saraf, kardiovaskular, limfatik, dan respiratori didedahkan. Fungsi dan hubungan antara sistem badan bagi pengekalan homeostasis badan dibincangkan

This course covers basic concepts of human anatomy and physiology. Basic tissues, integumentary, musculoskeletal, nervous, cardiovascular, lymphatic and respiratory systems are exposed. Functions and relationships among the human body systems to maintain homeostasis are discussed

- NUR3106 Anatomi dan Fisiologi II/ *Anatomy dan Physiology II* 3(3+0)
 Prasyarat : Tiada
 Kursus ini merangkumi konsep asas anatomi dan fisiologi manusia. Sistem endokrin, gastrousus, urinari, deria khas, dan reproduksi didedahkan. Fungsi dan hubungan antara sistem badan bagi pengekalan homeostasis badan dibincangkan
This course covers basic concepts of human anatomy and physiology. endocrine, gastrointestinal, urinary, special senses and reproduction systems are exposed. Functions and relationships among the human body systems to maintain homeostasis are discussed
- NUR3107 Asas Mikrobiologi, Parasitologi dan Imunologi/ *Basic Microbiology, Immunology and Parasitology* 2(2+0)
 Prasyarat : Tiada
 Kursus ini merangkumi asas mikrobiologi, parasitologi dan imunologi. Agen penyebab penyakit, patofisiologi dan tindakbalas sistem imun didedahkan. Patofisiologi dan manifestasi klinikal penyakit-penyakit akibat gangguan sistem imun dibincangkan
This course covers basics of microbiology, parasitology and immunology. Disease causative agents, pathophysiology and response of the immune system are exposed. Patophysiology and clinical manifestation of the diseases due to disturbance in the immune system are discussed
- NUR3108 Asas Hematologi dan Patologi/ *Basic Hematology and Pathology* 2(2+0)
 Prasyarat : Tiada
 Kursus ini merangkumi asas hematologi dan patologi. Fungsi sel darah, tindakbalas tisu terhadap kecederaan serta perkaitannya dengan penyakit dan rawatan dibincangkan. Neoplasia dan terapi transfusi juga didedahkan
This course consists the basics of hematology and pathology. The functions of blood cell, tissue response to injury and it's relationship to disease and treatment are discussed. Neoplasm and transfusion therapy are also exposed
- NUR3109 Asas Pemakanan Manusia/ *Basic Human Nutrition* 3(3+0)
 Prasyarat : Tiada
 Kursus ini meliputi asas pemakanan yang merangkumi biokimia dan fisiologi pemakanan manusia. Masalah ketidakseimbangan nutrisi dan pengambilan nutrien yang seimbang dibincangkan
This course covers basic nutrition that encompasses biochemistry and physiology of human nutrition. Nutritional imbalance problems and the intake of well balanced nutrients are discussed
- NUR3201 Konsep dan Amalan Kejururawatan I/ *Nursing Concepts and Practice I* 3(2+1)
 Prasyarat : Tiada
 Kursus ini meliputi konsep asas jagarawatan holistik dan respon fisiologi normal. Kemahiran jagarawatan asas ditumpukan. Komunikasi yang berkesan dalam jagarawatan klien diterapkan
This course covers the basic concepts of holistic nursing care and normal physiological response. Basic nursing care skills are focused. Effective communication in the nursing care of clients are applied

NUR3202 Konsep dan Amalan Kejururawatan II/ *Nursing Concepts and Practice II* 3(2+1)

Prasyarat : NUR3201

Kursus ini merangkumi konsep dan amalan dalam pemberian ubat, perawatan luka dan penjagaan klien operatif. Jagarawatan klien meliputi aspek biopsikososial ditekankan. Prinsip aseptik ditekankan dalam perawatan operatif dan luka

This course covers the concepts and practice of medication administration, wound care and the care of the operative client. Nursing care to client include the biopsychosocial aspects is emphasized. Principles of asepsis is focused in the care of operative client and wound care.

NUR3208 Penilaian Klinikal/ *Clinical Assessment* 3(2+1)

Prasyarat : Tiada

Kursus ini merangkumi kemahiran penilaian kesihatan klien. Penilaian kesihatan secara holistik dengan mengintegrasikan kemahiran komunikasi didedahkan. Perancangan jagarawatan klien berdasarkan tafsiran penemuan dalam penilaian ditekankan

This course encompasses health assessment skills of clients. Holistic health assessment integrated with communication skills is exposed. Nursing care plan of client is based on the health assessment findings are emphasized

NUR3211 Jagarawatan Medikal Surgikal I/ *Medical Surgical I* 3(2+1)

Prasyarat : Tiada

Kursus ini merangkumi jagarawatan klien dewasa yang mengalami gangguan sistem kardiovaskular dan respiratori. Patofisiologi dan perawatan penyakit dibincangkan. Jagarawatan klien dengan menggunakan proses kejururawatan ditekankan

This course encompasses the nursing care of adult clients with altered functions of the cardiovascular and respiratory. Pathophysiology and treatment of diseases are discussed. Nursing care of clients using nursing process are emphasized

NUR3212 Jagarawatan Medikal Surgikal II/ *Medical Surgical II* 4(3+1)

Prasyarat : Tiada

Kursus ini merangkumi jagarawatan klien dewasa yang mengalami gangguan dalam sistem hematologi, imunologi, nefrouinari dan gangguan keseimbangan cecair dan elektrolit. Patofisiologi dan perawatan penyakit dibincangkan. Jagarawatan klien dengan menggunakan proses kejururawatan ditekankan

This course encompasses the nursing care of adult clients with altered functions of hematology, immunology, nephrouinary, fluid and electrolytes imbalance. Pathophysiology and treatment of diseases are discussed. Nursing care of clients using nursing process are emphasized

NUR3213 Jagarawatan Medikal Surgikal III/ *Medical Surgical III* 3(2+1)

Prasyarat : Tiada

Kursus ini merangkumi jagarawatan klien dewasa yang mengalami gangguan sistem gastrousus, organ aksesori dan endokrin. Patofisiologi dan perawatan penyakit dibincangkan. Jagarawatan klien dengan menggunakan proses kejururawatan ditekankan.

This course encompasses the nursing care of adult clients with altered functions of gastrointestinal, accessory organ and endocrine. Pathophysiology and treatment of diseases are discussed. Nursing care of clients using nursing process are emphasized

NUR3214 Jagarawatan Medikal Surgikal IV/ *Medical Surgical IV* 3(2+1)

Prasyarat : Tiada

Kursus ini merangkumi prinsip asas jagarawatan kecemasan dan jagarawatan klien dengan gangguan fungsi pada sistem musculoskeletal dan neurologi. Patofisiologi dan perawatan penyakit dibincangkan. Jagarawatan klien dengan menggunakan proses kejururawatan ditekankan

This course encompasses the principles of basic emergency care and nursing care of clients with altered functions of musculoskeletal and neurology. Pathophysiology and treatment of diseases are discussed. Nursing care of clients using nursing process are emphasized

NUR3215 Jagarawatan Medikal Surgikal V/ *Medical Surgical V* 3(2+1)

Prasyarat : Tiada

Kursus ini merangkumi jagarawatan klien dengan gangguan fungsi pada sistem optalmik, otorinolaringologi, integumentari dan kanser. Patofisiologi dan perawatan penyakit dibincangkan. Jagarawatan klien dengan menggunakan proses kejururawatan ditekankan

This course encompasses nursing care of clients with altered functions of ophthalmic, otorhinolaryngology, integumentary system and cancer. Pathophysiology and treatment of diseases are discussed. Nursing care of clients using nursing process are emphasized

NUR3216 Penyakit Berjangkit/ *Infectious Diseases* 2(2+0)

Prasyarat : Tiada

Kursus ini merangkumi epidemiologi dan pencegahan penyakit berjangkit. Aspek penting dalam pengurusan dan perawatan klien di hospital dan komuniti dibincangkan

This course encompasses epidemiology and prevention of infectious diseases. Important aspects in the management and care of patients in the hospital and community are discussed

NUR3308 Perkembangan Peribadi dan Profesional/ *Personal and Professional Development* 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi etika, undang-undang, kemahiran pemikiran kritis dan reflektif dalam amalan kejururawatan. Nilai etika dan profesionalisme dalam pengurusan klien mengikut kod etika kejururawatan ditekankan

This course consists the ethics, law, critical thinking and reflection in the nursing practice. Ethical values and professionalism in managing clients following nursing code of professional conduct are emphasized

NUR3309 Pengurusan dan Pentadbiran Kejururawatan/ *Nursing Management and Administration* 2(2+0)

Prasyarat : Tiada

Kursus ini merangkumi pengurusan, pentadbiran dan kepimpinan dalam kejururawatan. Kepelbagaian sistem pengurusan dalam bidang kejururawatan dibincangkan. Pengurusan klien berintegrasi dengan ahli profesion kesihatan yang lain juga ditekankan

This course consists of management, administration and leadership in nursing. Various nursing management systems are discussed. Management of clients integrated with other health professionals are also emphasized

- NUR3310 Asas Psikologi dan Sosiologi/ *Basic Psychology and Sociology* 3(3+0)
- Prasyarat : Tiada
- Kursus ini merangkumi prinsip asas psikologi dan sosiologi yang mempengaruhi tingkah laku manusia dalam konteks kesihatan. Pemilihan pendekatan psikologi dalam memberikan perkhidmatan perawatan kepada klien dipraktikkan. Kaedah kaunseling bagi membantu klien untuk mengubah gaya hidup didedahkan
- This course consists basic principles of psychology and sociology in influencing human behavior in the health context. Selection of psychological approaches in providing treatment to the clients are practiced. Counseling methods for helping clients to change their life style are exposed*
-
- NUR3406 Perawatan Kesihatan Komuniti/ *Community Health Nursing* 4(3+1)
- Prasyarat : Tiada
- Kursus ini merangkumi pelbagai aspek kesihatan komuniti dan penjagaan kesihatan primer bagi memberi perawatan komprehensif kepada individu, keluarga dan komuniti. Penjagaan kesihatan ibu dan anak ditekankan. Pengetahuan dan kemahiran dalam perkhidmatan kesihatan keluarga di komuniti diaplikasikan
- This course encompasses various aspects of community health and primary health care to provide comprehensive care to individual, family and community. Care of maternal and child health are emphasized. Knowledge and skills in family health services in the community are applied*
-
- NUR3407 Promosi Kesihatan/ *Health Promotion* 4(2+2)
- Prasyarat : Tiada
- Kursus ini merangkumi konsep kesihatan, statistik kesihatan dan promosi kesihatan. Kesan masalah kesihatan persekitaran dan pekerjaan terhadap kesihatan manusia dibincangkan. Promosi kesihatan dan peranan jururawat dalam pencegahan penyakit ditekankan. Program intervensi kesihatan komuniti berasaskan penilaian kesihatan komuniti dilaksanakan
- This course encompasses the concepts of health, health statistic and health promotion. Effect of environmental and occupational health problem on human health are discussed. Health promotion and the role of the nurse in the prevention of disease are emphasized. Community health intervention programme based on community health assessment is implemented*
-
- NUR3503 Jagarawatan Kanak-Kanak Sakit/ *Care of the Sick Child* 3(2+1)
- Prasyarat : Tiada
- Kursus ini merangkumi jagarawatan kanak-kanak sakit. Penyakit yang lazimnya berlaku di kalangan kanak-kanak didedahkan. Penglibatan keluarga dan komunikasi berkesan dalam jagarawatan kanak-kanak ditekankan
- This course encompasses the nursing care of sick children. Common diseases among children are exposed. Family involvement and effective communication in nursing care of children are emphasized*

- NUR3504 Jagaan Klien Kritikal/ *Care of the Critical Client* 3(2+1)
 Prasyarat : Tiada
 Kursus ini merangkumi jagarawatan klien kritikal di unit rawatan rapi. Kerja berpasukan dalam perawatan klien kritikal secara holistik ditekankan. Penjagaan spesifik kepada klien yang menggunakan bantuan pernafasan secara mekanikal didedahkan
This course consists on the nursing care of in intensive care unit. Team work in the holistic care for critically ill clients are emphasized. Specific care to client with mechanical ventilation is exposed
- NUR3506 Obstetrik dan Ginekologi/ *Obstetric and Gynaecology* 3(2+1)
 Prasyarat : Tiada
 Kursus ini merangkumi perawatan obstetrik dan ginekologi. Jagarawatan holistik klien berpandukan etika dan profesionalisme kejururawatan ditekankan. Pendidikan kesihatan diintegrasikan dalam perawatan klien.
This course encompasses obstetrics and gynaecology care. Holistic care based on nursing ethic and professionalism is emphasized. Health education is integrated in the care of the client
- NUR3521 Jagarawatan Warga Emas/ *Care of The Older Person* 3(2+1)
 Prasyarat : Tiada
 Kursus ini merangkumi proses penuaan dan implikasi perubahan demografi terhadap penjagaan warga emas. Pelbagai isu berkaitan perubahan warga emas dan implikasi terhadap kesihatan warga emas dibincangkan. Pendidikan kesihatan diintegrasikan dalam perawatan klien
This course encompasses the process of ageing and implications of demography changes on care of the elderly. Various issues on the changes and its implications on the health of the elderly are discussed. Health education is integrated in the care of the client.
- NUR3523 Perawatan Kesihatan Mental/ *Mental Health Care* 3(2+1)
 Prasyarat : Tiada
 Kursus ini merangkumi konsep kesihatan mental. Perawatan klien dengan gangguan kesihatan mental meliputi intervensi terapeutik dan terapi alternatif dibincangkan. Aspek perundangan dan etika dalam perawatan klien ditekankan
This course encompasses concepts of mental health. Care of the clients with mental health disorders includes therapeutic interventions and alternative therapy are discussed. Legal and ethical aspects in the care of client are emphasized
- NUR3530 Jagarawatan Kritikal II/ *Critical Care I* 4(2+2)
 Prasyarat : NUR3406, NUR3407
 Kursus ini meliputi konsep penjagaan kesihatan ibu, anak berisiko, remaja dan dewasa serta jaminan kualiti dalam perkhidmatan kesihatan. Penjagaan ibu antenatal dan postnatal yang berisiko, kanak-kanak kurang zat makanan dan remaja ditumpukan. Program untuk remaja atau dewasa bagi kawalan penyakit dan promosi kesihatan dirancang
This course covers the concepts of maternal and child health care at risk, adolescent and adult as well as quality assurance in health services. The care of antenatal and postnatal mothers at risk, children with nutritional deficiencies and adolescent are focused. Program for adolescence or adult for disease control and health promotion is planned

Prasyarat : NUR3504

Kursus ini memberi penekanan terhadap perawatan kritikal kepada klien di unit rawatan rapi. Perkhidmatan kejururawatan yang prihatin kepada klien kritikal dan keluarganya ditumpukan. Jagarawatan klien kritikal yang memerlukan bantuan pernafasan mekanikal didedahkan

This course emphasizes on the care of the critical client in the intensive care unit. Caring nursing services to clients and the family are focused. Nursing care for critically ill clients who require mechanical ventilation support are exposed

NUR4911 Latihan Klinik I/ *Clinical Practice I*

3 (0+3)

Prasyarat : Tiada

Kursus ini merangkumi konsep asas jagarawatan holistik klien di hospital. Komunikasi terapeutik, kawalan jangkitan dan proses kejururawatan ditekankan. Pelajar perlu menjalankan tatacara kejururawatan secara professional dalam jagarawatan klien.

This course encompasses holistic nursing care in the hospital. Therapeutic communication, standard precaution and nursing process are emphasized. Students are required to perform nursing procedures professionally in the care of client

NUR4912 Latihan Klinik II/ *Clinical Practice II*

3(0+3)

Prasyarat : NUR4911

Kursus ini merangkumi konsep asas jagarawatan holistik klien di hospital. Cara pemberian ubat, prinsip aseptik, asas pengurusan luka serta penjagaan klien operatif ditekankan. Prinsip aseptik dalam penjagaan pelbagai jenis luka diaplikasikan

This course encompasses holistic nursing care in the hospital. Drug administration, principles of aseptic technique, basic wound management as well as operative client care are emphasized. The aseptic principle in the care of various types of wounds are applied

NUR4913 Latihan Klinik III/ *Clinical Practice III*

4(0+4)

Prasyarat : NUR4912

Kursus ini merangkumi perawatan klien dewasa yang menghadapi masalah kesihatan berkaitan gangguan sistem kardiovaskular, respiratori, hematologi, imunologi, nefroulinari dan gangguan keseimbangan cecair dan elektrolit. Pengetahuan proses patofisiologi dan reaksi tingkah laku yang bersesuaian dalam jagarawatan klien ditekankan

This course covers care of the adult clients with health problems related to altered cardiovascular, respiratory, hematology, immunology, nephroulinary and fluid and electrolytes imbalance. Knowledge of the pathological processes and exhibit appropriate behavioral in the nursing care of clients are emphasized

NUR4914 Latihan Klinik IV/ *Clinical Practice IV*

3(0+3)

Prasyarat : NUR4913

Kursus ini merangkumi prinsip asas jagarawatan kecemasan serta jagarawatan klien yang menghadapi gangguan fungsi dalam sistem gastrousus, organ aksesori, endokrin, muskuloskeletal, neurologi, obstetrik dan ginekologi. Pengetahuan proses patofisiologi dan reaksi tingkah laku yang bersesuaian dalam jagarawatan klien diaplikasikan

This course covers the basic principles during emergency and care of clients with altered functions of gastrointestinal, accessory organ, endocrine, musculoskeletal, neurology, obstetrics and gynaecology. Knowledge of the pathological processes and exhibit appropriate behavioral reaction in the nursing care of clients are applied

NUR4915 Latihan Klinik VI/ *Clinical Practice V*

5(0+5)

Prasyarat : NUR4914

Kursus ini merangkumi perawatan klien yang menghadapi gangguan fungsi dalam sistem optalmik, otorinolaringologi integumentari serta kanak-kanak yang sakit, klien kritikal dan perawatan kesihatan mental. Pengetahuan proses patofisiologi dan reaksi tingkah laku yang bersesuaian dalam jagarawatan klien ditekankan

This course covers the care of clients with altered functions of ophthalmic, otorhinolaryngology, integumentary system besides caring for sick children, critical client and mental health. Knowledge of the pathological processes and exhibit appropriate behavioral reaction in the nursing care of the client are emphasized

NUR4916 Latihan Klinik VI/ *Clinical Practice VI*

3(0+3)

Prasyarat : NUR4915

Kursus ini memberi tumpuan pada perawatan klien di komuniti dan warga emas. Jagarawatan yang bersesuaian dengan komuniti dan warga emas ditekankan. Komunikasi efisien dan penyelesaian masalah dalam menjalankan jagarawatan dipraktikkan

This course focuses on the care of client in the community and elderly. Suitable nursing care for community and elderly are emphasized. Efficient communication and problem solving in implementing nursing care are practiced

NUR4917 Latihan Klinik VII/ *Clinical Practice VII*

5(0+5)

Prasyarat : NUR4916

Kursus ini merangkumi skil pengurusan klien di wad. Kemahiran kepimpinan and fungsi jururawat terlatih dalam wad bersama ahli pasukan kesihatan yang lain didedahkan. Kemahiran jagarawatan untuk klien dengan masalah medikal, surgikal dan kanser diperkukuhkan

This course encompasses patient and ward management skills. Leadership skill and functions of a staff nurse in the ward with other health care team members are exposed. The student's ability in the care of medical, surgical and cancer client are strengthened

NUR4959 Disertasi Bacelor/ *Bachelor's Dissertation*

8

Prasyarat : NUR3004

Kursus ini merangkumi penyediaan cadangan projek penyelidikan, perlaksanaan dan analisis data dengan mengambil kira isu etika penyelidikan. Pelajar perlu menyediakan tesis penyelidikan dan membuat pembentangan lisan

This course covered the preparation of a research project proposal, its implementation, and data analysis with research ethics consideration. Students are required to prepare a thesis and make an oral presentation

Jabatan Sains Bioperubatan/ Department of Biomedical Sciences

SBP3002 Kaedah Penyelidikan Bioperubatan/ Biomedical Research Methodology 2 (2+0)

Prasyarat : Tiada

Kursus ini merangkumi konsep dan kaedah penyelidikan bioperubatan, penentuan saiz sampel, pengumpulan data dan rekabentuk eksperimen.

This course comprises the biomedical research concepts and methods, sample size determination, data collection and experimental design.

SBP3003 Statistik Bioperubatan/ Biomedical Statistics 2 (1+1)

Prasyarat : Tiada

Kursus ini merangkumi konsep asas dan inferens statistik, analisis data, pengujian hipotesis dan membuat kesimpulan berasaskan ujian statistik yang berkenaan.

This course comprises the basic concepts of statistics and statistical inferences, data analysis, testing of hypothesis, and making conclusions based on the respective statistical tests.

SBP3004 Pengurusan Haiwan Makmal/ Laboratory Animal Management 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi peraturan dan garis panduan penggunaan dan pengendalian haiwan makmal yang betul serta pengurusan perumahan dan penyakit berdasarkan aspek anatomi dan fisiologi haiwan makmal.

This course covers the legislations and guidelines of proper use and handling techniques of lab animals, the management of housing and common diseases in lab animals based on the animal's anatomical and physiological aspects.

SBP3005 Pengurusan Makmal/ Laboratory Management 2 (2+0)

Prasyarat : Tiada

Kursus ini merangkumi peraturan asas ketika berada di dalam makmal. Kursus ini memberi penekanan kepada kemahiran mengendalikan peralatan asas makmal.

This course divulges the concept of safe and effective laboratory management. The important elements in the organization, laboratory operation and related acts are also discussed.

- SBP3009 Kemahiran Asas Makmal/ *Basic Laboratory Skills* 2 (1+1)
- Prasyarat : Tiada
- Kursus ini merangkumi peraturan asas ketika berada di dalam makmal. Kursus ini memberi penekanan kepada kemahiran mengendalikan peralatan asas makmal.
- This course encompasses the basic rules and regulations in the laboratory. This course emphasizes on the skills in handling basic laboratory equipments.*
-
- SBP3010 Sains Makmal/ *Laboratory Science* 3 (2+1)
- Prasyarat : Tiada
- Kursus ini merangkumi pengenalan, pengendalian dan penyelenggaraan peralatan berteknologi tinggi mengikut prosedur operasi piawai di dalam penyelidikan bioperubatan.
- This course encompasses introduction, handling and, maintenance of high technology instruments according to standard operational procedure (SOP) in biomedical research.*
-
- SBP3011 Patologi/ *Pathology* 2 (1+1)
- Prasyarat : Tiada
- Kursus ini merangkumi konsep tindakbalas tisu terhadap kecederaan, perkaitannya dengan penyebab, patogenesis dan diagnosis beberapa penyakit penting. Teknik persampelan tisu, histologi, histokimia dan immunohistokimia dalam diagnosis dan penyelidikan juga dibincangkan.
- This course covers the concept of tissue reactions to injury, relationship between causes, pathogenesis and diagnosis of important diseases. Tissue's sampling techniques, histology, histochemistry and immunohistochemistry in diagnosis and research also will be discussed.*
-
- SBP3012 Hematologi/ *Haematology* 3 (2+1)
- Prasyarat : Tiada
- Kursus ini merangkumi pembelajaran mengenai proses dan faktor yang terlibat dalam pembentukan sel darah. Perbincangan merangkumi fungsi sel darah merah, sel darah putih, trombosit, sistem pembekuan dan trombosis. Tindakbalas sel darah dalam penyakit, pengelasan gangguan hematologi, ujian diagnosis makmal, amalan makmal yang selamat dan jaminan kualiti turut juga dibincangkan.
- This course covers the formation of blood cells and factors involved. The functions of red blood cells, white blood cells, thrombocytes, coagulation system, thrombosis and blood groups are discussed. The response of blood cells in diseases, classification of specific haematological disorders, laboratory diagnostic tests, safe laboratory practice and quality assurance are also discussed.*
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- SBP3013 Sains Transfusi/ *Transfusion Science* 2 (1+1)
- Prasyarat : SBP3012
- Kursus ini merangkumi pengetahuan asas tentang transfusi darah, kaedah flebotomi, kumpulan dan komponen darah. Ujian-ujian berkaitan dengan pretransfusi dan reaksi transfusi serta jaminan kualiti transfusi darah turut dibincangkan.
- This course encompasses basic knowledge on blood transfusion, techniques in phlebotomy, blood grouping and component. Tests required for pretransfusion and transfusion reaction as well as quality control of blood transfusion are also discussed.*

SBP3014	Sains Tingkahlaku/ <i>Behavioral Science</i>	3 (3+0)
Prasyarat : Tiada		
Kursus ini memperkenalkan prinsip asas sains tingkahlaku yang mempengaruhi tingkahlaku manusia terutamanya dalam konteks pekerjaan. Ia merangkumi perkembangan asas manusia dari peringkat bayi sehingga ke peringkat umur lanjut, asas psikologi yang berkaitan tingkahlaku individu serta kumpulan.		
<i>This course introduces the basic principles of behavioural sciences in influencing human behavior especially in the working context. This encompasses human development from the infancy to elderly stage related to basic psychology of individual and group behavior.</i>		
SBP3111	Fisiologi Manusia/ <i>Human Physiology</i>	4 (3+1)
Prasyarat : Tiada		
Kursus ini merangkumi segala aspek fisiologi manusia. Perkaitan fungsi organ dan proses fisiologi dalam mengekalkan homeostasis dibincangkan. Pentafsiran ujian makmal dari setiap fungsi organ juga ditekankan.		
<i>This course encompasses all the aspects of human physiology. The relationship of organ functions and physiological process in maintaining homeostasis are discussed. Interpretation of laboratory tests from each organ function is also emphasized.</i>		
SBP3116	Organisasi Manusia, Sistem Integumen dan Muskuloskeletal/ <i>Human Organization, Integumentary and Musculoskeletal Systems</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini merangkumi pengenalan kepada anatomi perubatan, organisasi tubuh dan homeostasis serta struktur, organisasi dan fungsi sistem integument, rangka dan otot.		
<i>This course encompasses the introduction to medical anatomy, body organization and homeostasis, the structure, organization and functions of the integumentary, skeletal and muscular systems.</i>		
SBP3117	Sistem Saraf dan Endokrin/ <i>Nervous and Endocrine Systems</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini merangkumi struktur, organisasi dan fungsi sistem saraf dan endokrin serta menghubungkan sistem tersebut dengan proses homeostasis badan.		
<i>This course encompasses the structure, organization and functions of the nervous and endocrine systems and relate the systems to the homeostatic mechanism in the human body.</i>		
SBP3118	Sistem Kardiovaskular, Limfatik dan Respiratori/ <i>Cardiovascular, Lymphatic and Respiratory Systems</i>	3 (2+1)
Prasyarat : Tiada		
Kursus ini merangkumi struktur, organisasi dan fungsi sistem kardiovaskular, limfatik dan respiratori serta menghubungkan sistem tersebut dengan proses homeostasis badan.		
<i>This course encompasses the structure, organization and functions of the cardiovascular, lymphatic and respiratory systems and relate the systems to the homeostatic mechanism in the human body.</i>		

SBP3119 Sistem Pencernaan, Urinari dan Pembiakan/ *Digestive, Urinary and Reproductive Systems* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi struktur, organisasi, fungsi, pengawalan sistem pencernaan, urinari, keseimbangan asid-bes dan pembiakan. Proses pembiakan daripada peringkat persenyawaan dan perkembangan sehingga kelahiran serta penyusuan juga dibincangkan.

This course encompasses the structure, organization, functions, regulation of the digestive, urinary, acid-base balance and reproductive systems. The reproductive processes from fertilization and development to birth as well as lactation are also discussed.

SBP3120 Radiobiologi/ *Radiobiology* 2 (2+0)

Prasyarat : Tiada

Kursus ini merangkumi segala sifat, pengesanan, pengukuran, pengenalanpastian dan kesan biologi bahan radioaktif dan sinaran mengion ke atas organisma. Faktor biokimia, persekitaran dan bahan kimia yang memberi kesan ke atas tindakbalas organisma terhadap sinaran dibincangkan. Kursus ini memberi kefahaman kepada pelajar dalam penggunaan secara selamat bahan radioaktif dan sinaran mengion untuk penyelidikan dan perubatan.

This course encompasses the properties, detection, measurement, identification and biological effects of radioactive materials and ionic radiation on organism. Biochemical, environment and chemicals factors affecting the organism towards radiation are to discussed. This course allows students to understand the safety use of radioactive materials and ionic radiation in research and medicine.

SBP3201 Asas Biokimia Perubatan/ *Basic Medical Biochemistry* 3 (2+1)

Prasyarat : Tiada

Kursus ini meliputi perubahan proses biokimia merangkumi struktur kimia, ciri dan aspek fungsi komponen dalam sel. Metabolisma dan tapak jalan pengawalan biomolekul juga turut dibincangkan.

This course covers the changes in biochemical processes, including the structural, chemical, properties and functional aspects of cellular constituents. The metabolism and regulation of biomolecule are also discussed.

SBP3202 Biokimia Klinikal/ *Clinical Biochemistry* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi pemahaman patogenesis penyakit serta perubahan parameter biokimia klinikal dalam diagnosis penyakit. Teknik, prinsip analisis sampel klinikal, interpretasi data dan pengesanan untuk diagnosis penyakit turut dibincangkan.

The course encompasses understanding of pathogenesis of diseases and changes in clinical biochemistry parameters for disease diagnosis. Techniques, analysis principles of clinical sample, data interpretation and validation for disease diagnosis are also discussed.

SBP3301 Farmakologi/ *Pharmacology* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi prinsip farmakologi dan kegunaan drug dalam rawatan pelbagai penyakit. Penekanan diberikan ke atas sistem biologi, mekanisme dan lokasi tindakan dan juga pelbagai penggunaan terapeutik, keracunan/ kesan sampingan pelbagai kumpulan drug yang digunakan di dalam perubatan.

This course covers the principles and basis of pharmacology and utilization of drugs for treating various diseases. Emphasis is on the effects of drugs on a biological system, mechanism and location of action, therapeutic uses and side-effects/adverse effects drug reactions used in medicine.

SBP3302 Toksikologi/ *Toxicology* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi terapeutik, mekanisme tindakan dan keracunan dan kestabilan 'drug'. Faktor-faktor kimia organik, biokimia, asas fisiologi dan farmakologi untuk kumpulan 'drug' dibincangkan. Penekanan adalah diberikan kepada penggunaan terapeutik, dan kesan spesifik 'drug'.

This course covers the usage of drugs as therapeutic agents, mechanism of action, toxicity and drug stability. Factors such organic chemistry, biochemistry and basic physiology and pharmacology of groups of drugs are discussed. Emphasis is given on therapeutic utilization and specific actions of drugs.

SBP3303 Toksikologi Molekul/ *Molecular toxicology* 3 (3+0)

Prasyarat : SBP3302

Kursus ini memberi pemahaman tentang prinsip penting toksikologi dan kesan racun terhadap sistem mamalia dalam keadaan sihat dan berpenyakit. Pelajar akan mempelajari teknik menganalisa dan mengenalpasti racun untuk diagnosis termasuk interpretasi data toksikologi.

This course provides understanding of important principles in toxicology and the effect of poisons in the mammalian system in healthy and diseased conditions. Students will learn techniques to analyse and identify the poison for the diagnosis including toxicology data interpretation.

SBP3403 Asas Mikrobiologi Perubatan/ *Basic Medical Microbiology* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi pengetahuan asas mikrobiologi dan teknik asas pengkulturan dan diagnosis mikrob. Perkembangan kejadian penyakit juga dibincangkan.

This course covers basic knowledge of microbiology and basic techniques in microbial cultivation and diagnosis. The development of diseases also will be discussed.

SBP3404 Parasitologi dan Entomologi Perubatan/ *Medical Parasitology and Entomology* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi patofisiologi, imunologi, diagnosis, kemoterapi dan pencegahan penyakit bawaan parasit. Bahagian entomologi merangkumi artropoda sebagai vektor di dalam transmisi penyakit, serangan dan kecederaan tisu. Kawalan, pencegahan dan diagnostik makmal akan juga dibincangkan.

This course encompasses the pathophysiology, immunology, diagnosis, chemotherapy and prevention of the parasitic diseases. The entomology section covers arthropods as vectors in disease transmission, their tissue invasion and injuries. Control, diagnosis and prevention will also be discussed.

SBP3408 Imunologi Asas/ *Basic Immunology* 2 (2+0)

Prasyarat : Tiada

Kursus ini merangkumi konsep asas imunologi, struktur dan fungsi molekul utama sistem imun serta pengawalaturan pengaktifan imuniti inat dan perolehan.

This course covers the fundamental concepts of immunology, structure and function of major molecules of the immune system and regulation of innate and adaptive immunity.

- SBP3409 Immunologi Lanjutan/ *Advanced Immunology* 2 (1+1)
 Prasyarat : SBP3408
 Kursus ini merangkumi butiran perubahan molekul semasa gerakbalas imun. Mekanisme imunopatogenetik penyakit imun, tolerans, keautoimunan dan imunologi tumor turut dibincangkan.
This course encompasses molecular events during immune response. Immunopathogenetic mechanisms of immune disease, tolerance, autoimmunity and tumour immunology are discussed.
- SBP3410 Kultur Sel dan Tisu/ *Cell and Tissue Culture* 3 (2+1)
 Prasyarat : Tiada
 Kursus ini memperkenalkan konsep asas pengkulturan sel dan tisu mamalia. Prosedur pengkulturan termasuk teknik aseptik, sterilisasi, analisa pertumbuhan, pembezaan, kematian dan kontaminasi turut bincangkan.
This course introduces the fundamental concepts of mammalian cell and tissue culture. Cell culturing procedures including materials, aseptic technique and sterilization, analysis of cell growth, differentiation, death and contamination are also discussed.
- SBP3411 Pengenalan Biologi Sel/ *Introductory Cell Biology* 2 (2+0)
 Prasyarat : Tiada
 Kursus ini merangkumi aspek molekul bagi proses sel utama yang menjadi asas kepada pengawalan fungsi sel dalam organisma multisel. Proses ini termasuk sintesis dan pusing ganti protein, pengedaran membran, kematian dan pembaharuan sel, motiliti sel, serta komunikasi antara sel dan persekitaran mikro, dengan penekanan kepada mekanisme dan peranan pelbagai struktur sel dan molekul dalam pengawalan proses tersebut.
The course covers the molecular aspects of main cellular processes that are essential for regulating cell function in multicellular organisms. These cellular processes include protein synthesis and turnover, membrane trafficking, cell death and renewal, cell motility, as well as communication between cells and their microenvironment, with emphasis on the mechanisms and the role of various cellular structures and molecules in the regulation of these processes.
- SBP3502 Genetik Manusia/ *Human Genetics* 3 (2+1)
 Prasyarat : Tiada
 Kursus ini merangkumi pengenalan kepada aplikasi prinsip asas genetik bagi kajian kesihatan manusia dan penyakit. Asas genetik manusia merangkumi genetik Mendel dan populasi serta aplikasi asas klinikal. Kepentingan dan implikasi penyakit genetik pada peringkat populasi dan keluarga turut dibincangkan.
This course encompasses an introduction to the application of basic genetic principles to the study of human health and disease. The basic human genetics include Mendelian and population genetics, and basic clinical applications. The importance and implications of genetic disease at population and individual families level are also discussed.
- SBP3503 Biologi Molekul Gunaan/ *Applied Molecular Biology* 3 (2+1)
 Prasyarat : Tiada
 Kursus ini merangkumi kegunaan biologi molekul, kaedah khusus termasuklah pengklonan gen,

manipulasi DNA dan analisis gen dan protein, teknik dan peralatan terkini dan isu semasa dalam bidang biologi molekul seperti sel stem dan transgenik.

This course encompasses the application of molecular biology, specific approaches including gene cloning, DNA manipulation and analysis of gene and protein, recent techniques and tools and current issues in molecular biology area such as stem cell and transgenic.

SBP3504 Teknologi Sitogenetik Gunaan/ *Applied Cytogenetics Technology* 3 (2+1)

Prasyarat : Tiada

Kursus ini meliputi teknik konvensional dan molekular sitogenetik yang digunakan sama ada didalam makmal klinikal atau kemudahan penyelidikan. Aspek praktikal dan kepentingan teknologi terutamanya sebagai alatan diagnosis didalam bidang perubatan dibincangkan.

This course comprises of conventional and molecular cytogenetics techniques used either in clinical laboratory or research facilities. Practical aspects and the importance of technologies particularly as a diagnostic tool in medical fields are discussed

SBP3505 Bioteknologi Perubatan/ *Medical Biotechnology* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi penghasilan semula kaedah dan aplikasi teknologi khas dalam bioteknologi perubatan untuk industri yang berkaitan farmaseutikal, diagnostik dan nutraseutikal termasuk isu berkaitan etika dan perundangan.

This course encompasses of methodology reproduction and application of special technology in medical biotechnology for the industry related with pharmaceutical, diagnostic and nutraceutical which includes ethical and legal issues.

SBP3507 Genetik Asas/ *Basic Genetics* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi genetik klasik dan molekular yang meliputi prinsip warisan dan alasan bagi variasi dalam semua organisma hidup. Ini termasuk prinsip Mendel, genetik populasi dan molekular, mekanisme penentuan dan pertalian seks, struktur DNA dan fungsi, aberasi dan mutasi kromosom.

This course covers the classical and molecular genetics, the principles of inheritance and the reasons for the variation in living organisms. These include the principles of Mendel, molecular, and populations genetics, mechanisms of sex determination and sex linkage, structure and function of DNA, chromosomal aberrations and mutations.

SBP3510 Bioinformatik Asas/ *Basic Bioinformatics* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi konsep dan amalan bioinformatik dalam mutasi genetik dan filogenetik dalam alam maya internet. Penekanan adalah kepada aspek teori dan praktikal menggunakan perisian lembut untuk sumber maklumat pangkalan data, menganalisis serta mengolah jujukan asid nukleik dan struktur protein.

This course encompasses bioinformatics concepts in genetic mutation, phylogenetic and practice that are available in the virtual environment of the internet. Emphasis will be given to the theoretical and practical aspects of software for the database resources, analysis and manipulation of nucleic acid and amino acid sequences.

- SBP3511 Informatik Kesihatan/ *Health Informatics* 2 (2+0)
- Prasyarat : Tiada
- Kursus ini meliputi aspek informatik kesihatan yang melibatkan organisasi, sistem dan pengurusan rekod. Isu berkaitan kesihatan awam seperti sistem kewangan, pentadbiran, jaminan kualiti dan etika turut dibincangkan.
- This course covers the aspects of health informatics which included organization, systems and record management. Issues related to public health such as system in financial, administration, quality assurance and ethics are discussed.*
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- SBP4301 Kimia Farmaseutikal/ *Pharmaceutical Chemistry* 3 (3+0)
- Prasyarat : SBP3301
- Kursus ini merangkumi penggunaan drug sebagai agen terapeutik, mekanisme tindakan, keracunan dan kestabilannya. Penekanan diberikan ke atas pelbagai faktor iaitu kimia organik, biokimia, asas fisiologi dan farmakologi, yang mempengaruhi penggunaan terapeutik dan kesan spesifik kumpulan drug berbeza.
- This course covers usage of drugs as therapeutic agents, mechanism of action, toxicity and drug stability. Emphasize is given on various factors namely organic chemistry, biochemistry, basic physiology and pharmacology, which influence the therapeutic utilization and specific actions of different groups of drug.*
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- SBP4302 Etnofarmakologi/ *Ethnopharmacology* 3 (3+0)
- Prasyarat : Tiada
- Kursus ini adalah pengenalan kepada etnofarmakologi, bidang pelbagai disiplin menggunakan pengetahuan kimia, ekologi, biologi dan farmakologi untuk menilai dan memahami kegunaan tumbuhan (dan sebatian lain) dalam sistem perubatan. Pelajar diperkenalkan kepada kesemua disiplin yang terlibat dalam penyelidikan etnofarmakologi dan menekankan persoalan berkaitan tindakan produk semulajadi ke atas badan manusia, keberkesanan tumbuhan ubatan dan pembangunan farmaseutikal berasaskan perubatan tradisional.
- This course is an introduction to ethnopharmacology, a multidisciplinary field of study that employs chemistry, ecology, biology, and pharmacology to evaluate and understand the use of plants (and other substances) in medical systems. Students introduced to all of the disciplines involved in ethnopharmacological research and address questions related to the actions of natural products in the human body, the efficacy of medicinal plants and the development of pharmaceuticals based on traditional medicines.*
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- SBP4401 Mikrobiologi Perubatan Lanjutan/ *Advanced Medical Microbiology* 3 (2+1)
- Prasyarat : SBP3403
- Kursus ini membincangkan hubungan mikroorganisma dan persekitaran dengan patogenesis penyakit, epidemiologi dan diagnosis makmal. Strategi kawalan serta pencegahan penyakit berjangkit turut dibincangkan.
- This course discusses the relationship of microorganisms and environment with pathogenesis of diseases, epidemiology and laboratory diagnosis. Control strategies and prevention of infectious diseases are also discussed.*

SBP4402	Pengisyaratan Sel dan Transduksi Isyarat/ <i>Cell Signaling & Signal Transduction</i>	2 (2+0)
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Prasyarat : Tiada

Kursus ini merangkumi ciri asas berbagai sistem pengisyaratan ekstrasel dan intrasel dan kepentingannya dalam berbagai sistem biologi. Ini termasuk komponen utama mekanisme pengisyaratan ekstrasel dan intrasel dan bagaimana komponen tersebut menghasilkan tapak-jalan pengisyaratan. Kursus ini juga memperkenalkan Cytoscape, satu perisian bioinformatik sumber terbuka bagi menghasilkan peta jaringan berbagai tapak-jalan pengisyaratan terpilih.

The course encompasses the basic features of various extracellular and intracellular signaling systems, and their importance in biological systems. This covers the key components of extracellular and intracellular signaling mechanisms and how these components create signaling pathways. The course introduces Cytoscape, an open source bioinformatics platform for creating network maps of selected signaling pathways.

SBP4501	Bioinformatik dan Sumber Bioperubatan/ <i>Bioinformatic and biomedical sources</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi penggunaan pengaturcaraan komputer untuk mengenalpasti kepelbagaian bio semula jadi dalam penyelidikan bioperubatan dan pengendalian data biologi. Aplikasi bioinformatik dalam merumus pemprosesan data umpamanya terapi gen dan perubatan molekul dibincangkan.

This course encompasses the use of computer programs for identification of natural diversity resources in biomedical research and handling of biological data. Bioinformatics application in data processing such as genes therapy and molecular medicine are also discussed.

SBP4502	Forensik Asas/ <i>Basic Forensic</i>	3 (2+1)
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Prasyarat : Tiada

Kursus ini merangkumi prinsip dan aplikasi ilmu sains dan teknologi bagi menganalisis pelbagai bukti dari tempat kejadian jenayah. Ia meliputi ujikaji makmal, pemerhatian peribadi, mencatat, membaca dan membuat pemerhatian terhadap filem yang berunsurkan pendidikan, dan komersial berkaitan dengan ilmu forensik. Keupayaan dan kelayakan pakar penyiasat, keadaan keputusan yang dikeluarkan dari makmal, dan undang-undang yang digunakan dalam memperakukan keputusan ujian dibincangkan.

This will cover the principles and the application of science and technology to various analyses of crime scene evidence. It includes experimentation, personal observation, note taking, reading and attention to educational films, and commercial films related to forensic science. Scope of expert investigation and qualification, the nature of the results which may be expected from the laboratory and the law as it applies to the admission of test results in evidence discussed.

SBP4504	Variasi dan Evolusi Manusia/ <i>Human Variation and Evolution</i>	3 (3+0)
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Prasyarat : Tiada

Kursus ini merangkumi variasi dan evolusi manusia dari aspek genotip dan fenotip individu dan populasi. Pendekatan semasa untuk analisis variasi dan evolusi manusia termasuk implikasinya turut dibincangkan.

This course encompasses genotype and phenotype aspects of human variation and evolution in individual and populations. Current approaches for analysis of human variation and evolution includes the implications are discussed.

SBP4601 Gangguan Genetik Manusia/ *Human Genetic Disorders* 3 (2+1)

Prasyarat : SBP3502

Kursus ini merangkumi hubungkait antara aberasi genetik dengan penyakit genetik manusia. Kesan variasi genetik, diagnosis dan rawatan yang bersesuaian juga dirangkumi. Teknik dan isu etika berkaitan dengan kajian genetik akan dibincangkan.

This course covers the relationship between genetic aberrations and various human genetic disorders. The effects of genetic variation, appropriate diagnosis and treatment are also included. Techniques as well as ethical issues related to genetic research will also be discussed.

SBP4602 Neurosains Perubatan/ *Medical Neuroscience* 3 (2+1)

Prasyarat : SBP3110/ SBP3111/ SBP3113

Kursus ini merangkumi prinsip neurosains berkaitan dengan perkembangan dan fungsi sistem saraf. Penyakit sistem saraf manusia dan model penyelidikan neurosains semasa dibincangkan.

This course covers the principles of neuroscience related to the development and functions of the nervous system. Diseases associated with human nervous systems and current neuroscience research models are discussed.

SBP4604 Sel Stem dan Perubatan Regeneratif/ *Stem Cell and Regenerative Medicine* 3 (2+1)

Prasyarat : Tiada

Kursus ini merangkumi pengetahuan asas saintifik sel stem di dalam perubatan regeneratif. Pelbagai proses pembezaan sel stem untuk aplikasi klinikal, perkembangan penyelidikan dan penemuan terkini teknologi sel stem beserta isu etika dan keselamatan dalam perubatan dibincangkan.

This course covers the basic scientific knowledge of stem cells in regenerative medicine. Various differentiation processes of stem cells for clinical applications, the development of research and the latest findings of stem cell technology together with safety and ethical issues in medicine are discussed.

SBP4904 Praktikum Bioperubatan/ *Biomedical Practicum* 4 (0+4)

Prasyarat : Tiada

Kursus ini mendedahkan pelajar kepada persekitaran pekerjaan sebenar, teknik dan tugas rutin yang dijalankan di makmal. Pelajar akan mempelajari pelbagai aspek dari pengumpulan sampel, analisis dan seterusnya mengesahkan keputusan. Kursus ini juga mendedahkan pelajar kepada kegunaan dan pengurusan peralatan selain dari memperoleh kemahiran asas dalam amalan makmal.

This course exposes the students to the real working environment, techniques and tasks routinely carried out in laboratories. Students will learn the various aspects of sample collection, analysis and result validation. The course also exposes the students to the use and management of equipments as well as gaining basic skills in laboratory practices.

SBP4905 Latihan Industri/ *Industrial Training* 8 (0+8)

Prasyarat : Tiada

Melalui kursus ini, pelajar didedahkan kepada persekitaran kerja yang sebenar di industri/organisasi. Latihan merangkumi aplikasi aspek teori dan praktikal yang telah dipelajari dengan amalan semasa di tempat kerja. Kemahiran menyelesaikan masalah dan komunikasi turut diberi penekanan.

In this course, students are exposed to real working environment in industries/organizations. Training includes application of the theoretical and practical aspects that have been studied with current practices in the workplace. Problem solving and communication skills are also emphasized.

SBP4959 Disertasi Bacelor/ *Bachelor Dissertation*

8 (0+8)

Prasyarat : Tiada

Kursus ini merangkumi penyediaan cadangan, pelaksanaan dan penulisan saintifik untuk sesuatu projek penyelidikan. Pendekatan saintifik bagi menjana data secara sistematik melalui rekabentuk, pengumpulan dan analisis data yang sesuai diberi penekanan.

This course covers the preparation of proposal, implementation and scientific writing of research project. Scientific approach to generate data systematically through appropriate design, data collection and analysis are emphasized.